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DENVER RTD EAGLE P3

Communication System

Software Requirements Specification

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1.0 Introduction

1.1 Purpose

This Software Requirement Specification (SRS) describes both the functional and non-functional requirements necessary to fulfill the technical specifications of the DENVER RTD Eagle P3 Communication System (CS) as detailed in “DENVER RTDTS(Conformed Version, March2010)”. This document shall be used by the project team members who will develop and verify the Communication System. If there’s no special mention, every requirement defined in this document has the priority.

1.2 Scope

The Communication System to be installed by WOOJIN within the DENVER RTD Eagle P3Carsisafullyintegrated Public Announcement(PA)and Transit Passenger Information System(TPIS). The system provides for the storage and management of operational traininformation,emergencyinformation,andpublicannouncementsthrougha fully integrated audio visual communications system. WOOJIN shall supply install test and commission all communication equipment necessary to meet the requirements of DENVER RTDTS(Conformed Version, March2010) Part C Rolling Stock Requirements.

1.3 Acronyms and Abbreviations

Acronyms & Abbreviations	Description
AAB	Audio Amplifier Board
AMF	Automated Message File
AMF	Audio Message Format
ANSM	ANS Microphone
AVL	Automated Vehicle Location System
CCP	Communication Control Panel
CMR	Cab Make-Up Relay
CNCS	Content & Control Server
CRA	Communication Rack A
CS	Communication System
DCV	Direct current Voltage
DMA	Driver Microphone Assembly
DPH	Driver PA Handset
DPO	Driver PA Operation Box
ESP	Exterior Speaker
FDI	Front Destination Indicator
GPS	Global Positioning System
ISP	Interior Speaker
LIC	LonWorks Interface Card
MDS	Monitoring and Diagnostic System
PA	Public Announcement
PAC	PA Control Board
PAC-MA	PAC - Married Pair Car A.
PAC-MB	PAC - Married Pair Car B.
PAS	Power Supply Assembly
PEI	Passenger Emergency Intercom
PID	Passenger Information Display
PII	LED Passenger Information Indicator
PIS	Passenger Information System
PTU	Portable Test Unit
SDI	Side Destination Indicator
SDI	Serial Digital Interface
TRF	Transformer
UDP	User Datagram Protocol
VOIPC	Voice over Internet Protocol Card
VRC	Voice Recording Card
VRX	Video RX Board
VTX	Video TX Board
WLAN	Wireless LAN
WLANS	Wireless LAN System
WLR	Wireless LAN Router

1.4 Reference Documents

Publisher	Document Number	Title
DENVER	18FH012	DENVER RTDTS(Conformed Version, March2010) Part C Rolling Stock Requirements
Rotem	REDU106113	Procurement Technical Specification For Denver RTD EMU Communication System
IEEE	830-1998	ANSI/IEEE Standard 730, 828, 830, 1012A, 1016, 1228

1.5 Structure of Document

The remainder of this SRS is divided into two main sections:

Section 2: General Description

This section describes the hardware and software context of this particular communication, as well as its intended operations, functions and constraints. This section does not specify the requirements for the communication system, but provides the background information necessary to interpret the requirements.

Section 3: Specific Requirements

This section provides details of all of the specific requirements for the CS. The document is organized in a manner consistent with and in a corresponding order to the Master Software component list as described in Section 7 of the WOOJIN SFD document 423-309. Each software component is addressed separately regarding its external interfaces and functional requirements.

The following categories:

- External Interfaces
- Functional Requirements
- Performance Requirements
- Database Requirements
- Design Constraints
- Software System Attributes

2.0 General Description

2.1 Product Perspective

- The WOOJIN CS provides for all communication functionality within the newly fabricated DENVER RTD Eagle P3 cars. The system manages all aspects of audio, video, and data communications pertaining to passenger information and operational information relative to the vehicles and their interaction with DENVER RTD Wayside & Control Network. The context diagram shown in section 2.1.1 below, (System Interfaces), illustrates the various interface points between the CS and external entities.

- All CS application S/W components running on the onboard CNCS are operating under the Windows XP embedded OS platform. Device specific software components are written as low level code utilizing the inherent instruction set of the specific Micro processor controlling the device. There are also a number of COTS software components utilized within the Wayside and On-Board systems.

- The WOOJIN CS software components utilized do not require any ongoing updating, unless there are changes made to the H/W configuration. In the event that additional functional requirements are specified in the future, they may result in the need for S/W modifications to specific components.

2.1.1 System Interfaces

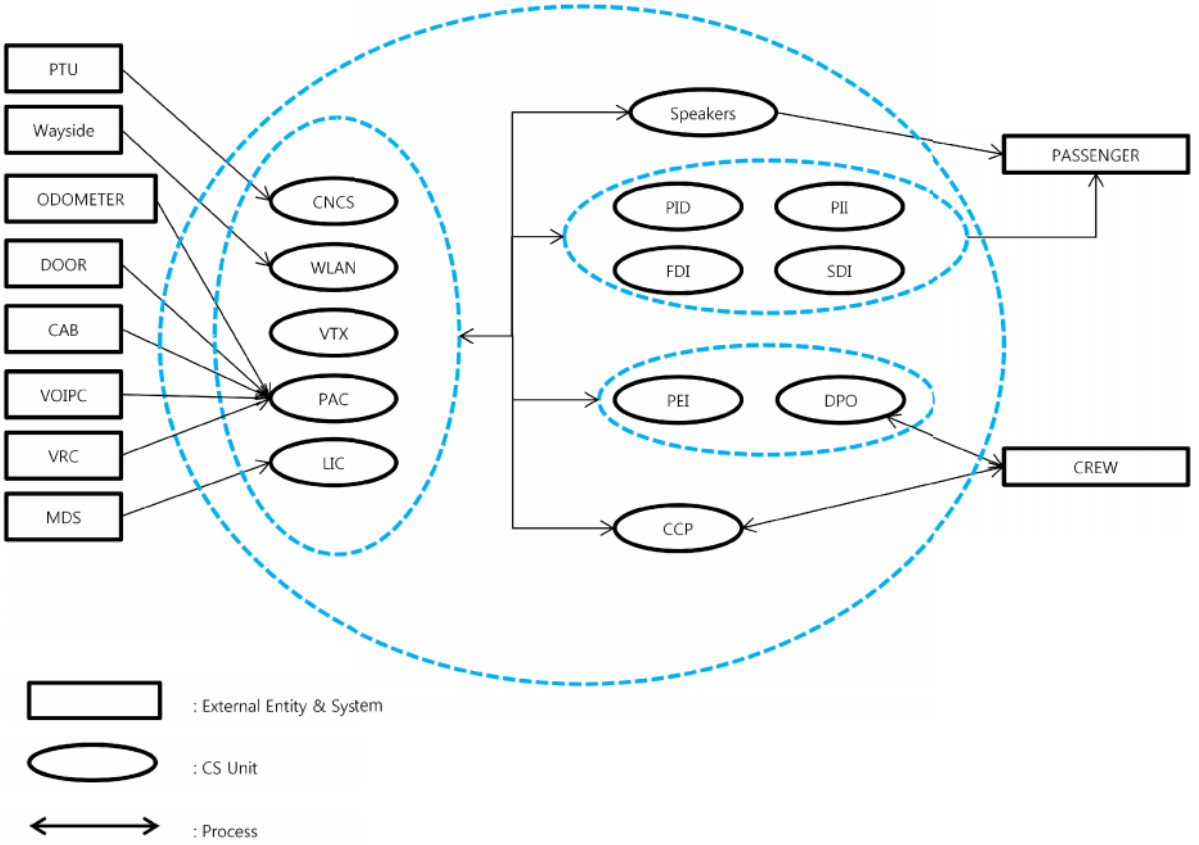


Figure 2.1.1-A. Communication System Context Diagram

From a software overview perspective, the CS as a functional entity interacts with various internal and external entities as illustrated in Figure 1 above. To accomplish this interaction, the CS relies on a range of user, hardware, software, and communications interfaces to provide for the required DENVER RTD specified functionality. The following subsections describe the various interface categories, the specific entities to be communicated with, the nature of the communication, and the physical devices involved.

2.1.2 User Interfaces

There are four user types that require human interfacing to the communication system. Each of these entities can function as either the source or destination of information processed and managed by the system.

External Entity	Description
Driver	CAB Speakers, Driver Mic., CCP Touch Screen, DPO Box
Conductor	PA Speakers,
Passenger	LED, LCD, Internal & External Speakers, PEI Box,
Maintenance	CCP, PTU

2.1.3 Hardware Interfaces

External Entity	Description
MDS	The MDS interfaces with LIC using LonWorks Interface protocol.
MASCON Key Switch	The CCP and PAC interface with the MASCON Key via a hard wired signal line utilizing a DCV input.
Access door Equipment	The PAC interfaces with access door equipment via a hard wired signal line utilizing a DCV input.
PTU	The CNCS interfaces with the PTU via the build in USB / RS232 port located on the front of the CS Rack
Other Cars	The PAC's of each individual Car interfaces with the other cars of the consist via the use of RS-485 over coupled train lines.

2.1.4 Software Interfaces

External Entity	Description
MDS Data Collection Node	The onboard CNCS communicates with the several MDS Data Collection Nodes located at the Yards and Union Station. This software interface accommodates the automated process of downloading MDS data from the Onboard communication system to the Wayside MDS data collection nodes.
Data Upload	This interface represents an automated process by which data is uploaded to the Onboard systems from the Wayside. This interface consists of a predefined set of processes which are governed by a set of operational parameters. This interface is used to provide for maintaining current versions of software, schedules, and content.

2.1.5 Memory Constraints

Reflect the design not to use the usable memory more than 80% when designing CNCS S/W.

Sub System	Memory Type	Size	Description	Use	Usable Memory
CCP	Flash	2Gbyte	NAND Flash RAM incorporated into the microcontroller	Execute Code	970Mbyte
	Ram Usable memory	256Mbyte	RAM incorporated into the microcontroller	Variable, Stack, Working	59Mbyte
PAC	On-Chip Flash	256Kbyte	Flash ROM incorporated into the microcontroller	Execute Code	156Kbyte
	On-Chip RAM	64Kbyte	RAM incorporated into the microcontroller	Variable, Stack, Working	32Kbyte
	Off-Chip RAM	1Mbyte	Flash RAM accessed through the Microcontroller Bus	Configuration data	90Kbyte
VOIPC	Flash	1Gbyte	NAND Flash RAM incorporated into the microcontroller	Execute Code	970Mbyte
	Ram Usable memory	91Mbyte	RAM incorporated into the microcontroller	Variable, Stack, Working	59Mbyte
VRC	External RAM	2Gbyte	RAM accessed through the Microprocessor Bus (DDR2 SDRAM)	OS, Execute Code, Working	1Gbyte
LIC	On-Chip Flash	256Kbyte	Flash ROM incorporated into the microcontroller	Execute Code	200Kbyte
	On-Chip RAM	64Kbyte	RAM incorporated into the microcontroller	Variable, Stack, Working	32Kbyte
FDI, SDI PII	On-Chip Flash	64Kbyte	Flash ROM incorporated into the microcontroller	Execute Code	32Kbyte
	On-Chip RAM	16Kbyte	RAM incorporated into the microcontroller	Variable, Stack, Working	3Kbyte

2.1.6 Operational Assumptions

- The Onboard Communication System must be designed to remain fully functioning at all times during normal operations of the vehicle.
- Regardless of each software component's specific function, all components must operate in a synergistic manner consistent with reasonable expectations for a completely integrated communications system.
- All software components (as applicable) must be designed in a manner which supports the collection of system fault data and its subsequent logging into the MDS data terminal.
- PA functionality must include the ability to automatically mute the exterior speaker(s) audio level for specified stations at predetermined times.
- The Train Radio sub-system will be replaced with the VOIP function. Manual PA announcement functionality must remain active in the event of power loss for a period of 90 minutes.
- The priority order for functional use of the Communication System should follow the constraints of section 2.5.

2.1.7 Site Adaptation Requirements

None.

2.2 Devices with S/W Functions

The on-board Communication System utilizes a combination of application software, device firmware, and logic circuitry to provide for the required functionality. The following paragraphs describe each CS device that utilizes application or firmware software, as well as a general description of the type of functionality each sub-component provides. A complete detailed functional description of each software component is included in section 3.1.

2.2.1 PAC

The CS includes two distinct PA Controllers which are housed in both CRA. These devices provide for the physical aspects of managing and distributing both Audio and Control Data within a specific vehicle, as well as amongst the various vehicles making up a Train Consist. The PAC's process and distribute all audio messaging in accordance with the parameters specified within the AVL messaging system, the Intercom system, and the PA system. PAC Control software provides the following functions;

The PAC manages all audio messages such as AVL, IC (Intercom), PEI, PA Announcement and also distributes them to the appropriate devices.

The Master PAC stores the pre-recorded MP3 audio contents for Automatic PA announcements.
The individual PAC's monitor the status of all major Communication System devices and aggregates the status data at the Master PAC.

The Master PAC has a direct connection to the Master CCP and provides much of the information that is used by the CCP to display information to the crew.

The PAC's manage all RS-485 communication both within the Cars and across the Consist.
This includes performing the Consist Inauguration process.

2.2.2 VOIP

VOIPC(Voice Over Internet Protocol Card) enables the voice communication between the intercom and the OCC. VOIPC transmits voice as the OCC by using the VOIP protocol after it converts an intercom voice signal received to digital data. It converts the digital data received through the VOIP protocol from the OCC, then transmits it to the intercom device in the vehicle. The commands between the OCC and the intercom are controlled by using ton signals via VOIP.

2.2.3 VRC

VRC(Voice Recording Card) is the device that records communication contents of OCC to DPO and OCC to PEI. VRC stores communication dates, time and contents, then transmits them to CNCS by using FTP at the appointed time.

2.2.4 DPO

DPO is an intercom communication device which driver is able to use. It receives the input signal with tone signal and transmits output signal with tone. DPO is able to respond to the call by OCC, PEI communication request. 3-way communication with the OCC is available during the communication if required. If the DPO cannot respond to the PEI call, OCC will respond instead of the DPO.

DPO consists of 7 buttons.

- Make an announcement by selecting the range of the manual PA announcement
- Communication between OCC and DPO, Multi-communication between DPO and other DPOs.
- Communication between DPO and PEI
- Initialize the call between intercoms and the call between PEIs.
- Communication between DPO and OCC, Make a call when 3-way communication with the OCC is required during communication between DPO or PEI.

2.2.5 DPH

DPH is a device which driver is able to make manual announcement. It has employed the manual announcement function separately from the DPO.

DPH consists of 3 buttons.

- Make an announcement by selecting the range of the manual PA announcement

2.2.6 PEI

PEI is a device which passenger is able to communicate with the driver or crew in an emergency situation. The location of the pressed PEI will be announced through the driver's speaker when passenger presses the CALL button, and communication will be available by composing communication path when PEI button has been pressed on the DPO.

2.2.7 CCP

The Communication Control Panel is the primary interface between the onboard CS and the Driver. This device is mounted in the Cab and is easily accessible to the Driver. It is equipped with a touch screen and GUI user interface which provides the crew with the ability to initialize the Consist, set & edit operating parameters, test functionality, review real time schedule adherence data, and monitor the CS operational status. The CCP performs the process of calculating the Train's current position as compared to the defined route schedule which support both the Schedule Adherence and AVL automated message triggering functions.

2.2.8 CNCS

The Communication and Control Server represents the heart of the onboard communication system. It is housed in CRA and is equipped with an Intel dual core processor, 2GB Ram, a 32GB solid state drive and operates under the Microsoft Windows XP Embedded OS. The CNCS provides for the storage and execution of many of the onboard CS software components, as well as for the storage and distribution of

most of the visual content. The following lists the various software elements which are sub-components of the CNCS and the functionality provided by each;

1) Launcher S/W Component (Launcher.exe): This application is automatically executed at the conclusion of the boot process. The primary functions of this software are;
Performs version control comparisons of newly downloaded program updates and sets update flags as appropriate.
Performs the application updates as set by the update flags.
This process is only executed once at power up, and then loads the "Main.exe" application upon completion.

2) Main Control S/W Component (Main.exe): This is the Content & Control Server's (CNCS) primary software component and it resides on the server's internal hard drive. After the CNCS has been initialized and the "Launcher.exe" application has completed all component update functions, then the CNCS automatically loads "Main.exe". Main.exe provides for management, storage and communications between and amongst the subsystems, as well as provides for the delivery of content parameters as defined in the scheduler. This application also spawns various other memory resident service applications and provides for communication between the various services utilizing UDP. The majority of all devices within the communication system are in some way related to, driven by, managed by, or dependent upon functionality provided for by this application.

3) Primary Media Player S/W (PP.exe): The Primary Media Player controls the main (upper) video playback zone of the LCD display devices. It receives content scheduling information provided by the CNCS's Main Control software application and facilitates the playback of specified content within the defined zone of the specified output devices.

4) Secondary Media Player S/W (SP.exe): The Secondary Media Player controls the lower (scroll) video playback zone of the LCD display devices. It receives content scheduling information provided by the CNCS's Main Control software application and facilitates the playback of specified content within the defined zone of the specified output devices.

5) FTP Data Update S/W (FTPData.exe): This software component is used to provide for both the uploading and downloading of content and data files between the on-board system and the Wayside network. This software utilizes FTP protocol to manage the transfer of the files over the Wayside's Wireless LAN. The software will employ the use of a process script to establish the priority order and manner by which it determines the specific files to be transferred, the naming conventions to be utilized, when they are to be transferred, and where they are to be placed. This software will continually monitor the availability of the Wayside LAN, will automatically connect to the LAN and determine the location of access, and will then perform any currently required file transfers.

6) MDS Data Update S/W (MDSData.exe): This software component provides functionality related to
a) transferring MDS data from the MDS to the CNCS utilizing the LonWorks interface, and
b) transferring CS software version information and GPS time.

7) Content Update S/W (ContentUpdate.exe): This software component provides functionality related to the automatic version control, updating and management of content and schedule files within the on-board communication system (AMF(Automated Message File)**,** MP3, Schedule, Play List, etc.). It manages the distribution of these content components from the CNCS to the internal memory modules of the various intended devices. This software utilizes the RS-485 Car Lines to distribute LED content to the displays, and to distribute the schedule file components to the CCP. It utilizes USB to distribute the MP3 audio files to the PAC.

2.2.9 VTX

VTX Board receives video signal from the CNCS and transmits it to the VRX Board. CNCS transmits video signal to the VTX through LVDS interface. VTX receives LVDS signal and outputs to the VRX by transferring to the SDI(Serial Digital Interface) signal which remote transmission is available.

2.2.10 VRX

VRX Board displays video of PID1 and PID2 devices by receiving video signal from the VTX. VTX Board transmits SDI(Serial Digital Interface) video signal which remote transmission is available to the VRX, and the VRX drives LCD of PID1 and PID2 by transferring SDI signal to the signal which is able to drive the LCD panel. Then it plays (3R , Regeneration/Retiming/Reamplication) SDI signal with PID1 and PID2.

2.2.11 FDI

The Front Destination Indicator (FDI) Control S/W component provides for the inherent functionality of the LED display device, as well as the functionality enabling the device to display Route# & Destination information. The S/W also facilitates RS-485 communication with the PAC controller to provide for content updating, content management, and system diagnostics.

It provides the control functionality for the actual LED Display.
It manages the receipt and execution of messaging parameters, content, and control codes.
It manages all RS-485 communication between the FDI's and the PAC's
It provides operational status information to the PAC via the RS-485.

2.2.12 SDI

The Side Destination Indicator (SDI) Control S/W component provides for the inherent functionality of the LED display device, as well as the functionality enabling the device to display Route# & Destination information. The S/W also facilitates RS-485 communication with the PAC controller to provide for content updating, content management, and system diagnostics.

It provides the control functionality for the actual LED Display.
It manages the receipt and execution of messaging parameters, content, and control codes.
It manages all RS-485 communication between the SDI's and the PAC's
It provides operational status information to the PAC via the RS-485.

2.2.13 PII

The Passenger Information Indicator (PII) Control S/W component provides functionality consistent with the FDI and SDI software but also monitors the AVL messaging system commands to display the automated information notifications.

It provides the control functionality for the actual LED Display.
It manages the receipt and execution of messaging parameters, content, and control codes.
It manages all RS-485 communication between the PII's and the PAC's
It provides operational status information to the PAC via the RS-485.
It monitors the AVL triggers and control signals to facilitate automated messaging on the PII's.

2.2.14 LIC

The LonWorks Interface Card (LIC) provides for the interface between the MDS and the CS. The LIC is inserted into the backplane of CRA, utilizes RS-232 to communicate with the CNCS, LonWorks to communicate with the MDS, and RS-485 to communicate with the PAC. The card has two distinct processor chips, which each having firmware elements as described below.

2.3 User Characteristics

User	Description
Crew	- Communicates with OCC through Driver Microphone and CAB Speaker. - Broadcasts public announcements of guidance to passengers using Microphone. - Operates and manages Communication System using Touch Screen.
Passenger	- Receives route guidance information and public announcements through LED and LCD. - Receives manual announcements of guidance from crew and automatic announcements of guidance from Communication System through Internal & External Speakers - Receives direct manual announcements of guidance from OCC as occasion demands

2.4 Constraints

- All application S/W operating on CNCS and CCP shall be created with Visual C++ 2008.
- Neuron C Language and Node Builder v3.1 compiler shall be used as the programming language for Lon Works Communication.
- IAR EWARM 4.42a, IARAVR 5.2b, KeiluVision2 and KeiluVision3 shall be used as the programming tool for microprocessors.

2.5 Assumptions and Dependencies

Public address Priority Order: The Public Address system utilizes the following priority order:

- Manual PA announcement OCC
- Manual PA announcement driver.
- Intercom communication Pre recorded Emergency PA announcement
- Pre recorded Automated PA announcement

All communication devices shall be available when the CAB-ON state is indicated. The Communication System should be fully functional during normal train operations.

Any manual announcement shall be train lined to all DENVER RTD cars within the Consist, and be routed to the external and/or internal speakers as appropriate. The Crew will utilize the CCP located in the Master Cab to configure and monitor the CS.

2.6 Apportioning of Requirements

None.

3.0 Specific Requirements

3.1 External Interface

1) CAB ON

Signal Name	CAB ON
Type	Digital Input
Description	Each PAC utilizes the adjacent Cab's CABON signal to determine if it should be set as the Master PAC for the entire Consist. The CAB ON signal will be derived from the Cab Make-Up Relay (CMR) by monitoring the output voltage of the CAB ON signal line. If the CAB ON signal is present, then the specified PAC will be designated as the Master PAC. As such it will manage and control the RS485 Train Line, perform the Train Consist configuration process, and manage all audio functionality related to the audio Train Lines.

2) LEFT DOOR OPEN (LDO)

Signal Name	LEFT DOOR OPEN (LDO)
Type	Digital Input
Description	The PAC utilizes this signal to determine if the left side door is in the Open or Closed position. The Door Control Unit (DCU) is equipped with a DCV output signal to indicate if the door is open or closed. The signal is activated just prior to the door beginning to open, and is terminated just after the door completes closing.

3) RIGHT DOOR OPEN (RDO)

Signal Name	RIGHT DOOR OPEN (RDO)
Type	Digital Input
Description	The PAC utilizes this signal to determine if the right side door is in the Open or Closed position. The Door Control Unit (DCU) is equipped with a DCV output signal to indicate if the door is open or closed. The signal is activated just prior to the door beginning to open, and is terminated just after the door completes closing.

4) FORWARD

Signal Name	FORWARD
Type	Digital Input
Description	The Master PAC receives the FORWARD signal from the Traction Control System.The FORWARD signal is provided by the Traction Control System whenever the MASCON Key is turned to the "Forward" position.

5) REVERSE

Signal Name	REVERSE
Type	Digital Input
Description	The Master PAC receives the REVERSE signal from the Traction Control System.The REVERSE signal is provided by the Traction Control System whenever the MASCON Key is turned to the “Reverse” position.

6) RS-485 CAR LINE

Signal Name	RS-485 CAR LINE
Type	Digital Input / Output
Description	The RS-485 Car Line data Input & Output signal is used to provide for digital communication between the PAC’s and the various devices within a Car. The RS-485 Car Line provides for data communication between the Primary PAC, CCP, Display Signs, CNCS,LIC, and the secondary PAC in the same car. Within any given car the Primary PAC maintains protocol control of the RS-485 Car Line network. All other RS-485 devices within the Car will typically be polled within a continuous loop by the Primary PAC. In most cases the primary PAC will issue requests for specific responses from each device, and the addressed device will provide the requested response. In certain cases it will be necessary for the Primary PAC to temporarily pass RS-485 control to another device so that it may affect the desired data communication.

7) RS-485 TRAIN LINE

Signal Name	RS-485 TRAIN LINE
Type	Digital Input / Output
Description	The RS-485 Train Line data I/O signal is used to provide for digital communication between the Master PAC and all slave PAC’s throughout a Consist. The Master PAC maintains protocol control of the RS-485 Train Line network, and polls all other PAC’s within the Consist utilizing are polled via a continuous loop by the Master PAC. The Master PAC either a) issues information telegrams addressed to specific PAC’s, or b) issues requests for specific responses from slave PAC’s. The slave PAC’s will either receive and store the data, or respond to a request by providing the requested information. In certain cases the Master PAC may temporarily pass RS-485 control to a slave PAC to perform a specific function, which will then return control after the function is complete.

3.1.1.1-A: RS-485 Polling Table

RS-485PollingSequences& Times

Train Line Poll Table

Equipment	Remarks	Access Time
PAC(Primary)	Cab On Car	0
PAC(Primary)#1	Other Car	200 ms
PAC(Primary)#2	Other Car	200 ms
PAC(Primary)#3	Other Car	200 ms
PAC(Primary)#4	Other Car	200 ms
PAC(Primary)#5	Other Car	200 ms

Ca rLine Poll Table

Equipment	Remarks	Access Time
CCP	CCP1~ 2	500m sec(Slave is about 14sec)
LIC	LIC1~2	2sec
CNCS	CNCS1~2	2sec
VOIPC	VOIP1~2	
VRC	VRC1~2	
PAC	PAC (Secondary)	Slaveisabout14sec
VTX	VTX	about 14sec
PID	PID1~PID12	about 14sec
FDI	FDI1~FDI2	about 14sec
PII	PII1~PII4	about 14sec
SDI	SDI1~SDI8	about 14sec

Device Statu sof Train Line

Equip.	Status Info.	Interface	Remarks
PAC	DPO/PEI/CCP /ETC.	Train Line RS-485	PACS tatus of All Train

Device Status of Car Line

Equip.		Status Info.	Interface	Remarks
CCP		Comm. State/ Device state	Car Line RS-485	Primary& Secondary
CR	LIC			Primary &Secondary
	PAC			Primary &Secondary
	VTX			Primary &Secondary
	CNCS			Primary &Secondary
	VOIPC			Primary &Secondary
	VRC			Primary &Secondary
PID				
FDI				
SDI				
PII				
DPO		Fault	Digital Output	*When Transmit to Mast PAC *Train Line 485
PEI				

3.1.1.1-B: RS-485 Address Table

Denver Comm. Internal485 ID Table Item								
1	CRA							
	LIC1	VOIPC1		VTX1	CNCS1	PAC1	VRC1	
	0x11	0x12		0x14	0x15	0x16	0x17	
	LIC2	VOIPC2	GIA2	VTX2	CNCS2	PAC2	VRC2	
	0x21	0x22	0x23	0x24	0x25	0x26	0x27	
2	CCP1	CCP2						
	0x31	0x32						
3								
4	PII1	PII2	PII3	PII4				
	0x51	0x52	0x53	0x54				
5	FDI1	FDI2						
	0x61	0x62						
6	SDI1	SDI2	SDI3	SDI4	SDI5	SDI6	SDI7	SDI8
	0x71	0x72	0x73	0x74	0x75	0x76	0x77	0x78
7	PID1-1	PID1-2	PID1-3	PID1-4	PID1-5	PID1-6	PID1-7	PID1-8
	0x81	0x82	0x83	0x84	0x85	0x86	0x87	0x88
8	PID2-1	PID2-2	PID2-3	PID2-4				
	0x91	0x92	0x93	0x94				

Denver Com. Train line 485 ID Table

Number	Train Number	Item	Train line 485 ID
1	Car1	PAC1(MA)	0x1
		PAC2(MB)	0x1
2	Car2	PAC1(MA)	0x2
		PAC2(MB)	0x2
3	Car3	PAC1(MA)	0x3
		PAC2(MB)	0x3
4	Car4	PAC1(MA)	0x4
		PAC2(MB)	0x4
5	Car5	PAC1(MA)	0x5
		PAC2(MB)	0x5
6	Car6	PAC1(MA)	0x6
		PAC2(MB)	0x6
7	Car7	PAC1(MA)	0x7
		PAC2(MB)	0x7
8	Car8	PAC1(MA)	0x8
		PAC2(MB)	0x8
9	Car9	PAC1(MA)	0x9
		PAC2(MB)	0x9
10	Car10	PAC1(MA)	0xA
		PAC2(MB)	0xA
11	Car11	PAC1(MA)	0xB
		PAC2(MB)	0xB
12	Car12	PAC1(MA)	0xC
		PAC2(MB)	0xC

8) COUPLER

Signal Name	RS-485 TRAIN LINE
Type	Digital Input
Description	Each PAC utilizes this signal to determine if the specific coupler which is directly adjacent to itself is in either the closed (coupled) or open (uncoupled) state. The COUPLER signal is a DCV hardwired signal which is generated by the Train Control system when a coupler device is in an open (uncoupled) state. During the Consist inauguration process, each PAC will evaluate the state of this signal to determine if another Car (2 PAC's) is attached to the coupler, or if the Car in which the PAC is located is the last Car of the Consist. Each PAC must evaluate the state of the attached coupler at a minimum rate of once every 500ms. The stored parameter value will be updated to reflect the current state when the physical Coupler is switched from the Open to Closed position, and vice versa.

9) IC LIVE Lines

Signal Name	IC LIVE Lines
Type	Digital Input
Description	Each PAC has 12 hardwired IC LIVE signal lines which are directly connected to two DPO's, two DPH's, four PEI control boxes. Each of these IC Devices has a distinct IC LIVE line to indicate if the device is powered on or not. If a DCV signal is present, then the device is powered on and is ready for operation. This signal does not indicate that a PEI Call is in progress, but merely indicates that the device is powered on and is operational.

10)DPO ADDRESS Lines

Signal Name	DPO ADDRESS Lines
Type	Digital Input
Description	Each PAC receives input from a discrete DPO ADDRESS signal line which is hardwired to the PAC from the DPO box located in the adjacent Cab. This ADDRESS line is used to determine if the DPO device has been activated by the driver.

11)DPH ADDRESS Lines

Signal Name	DPH ADDRESS Lines
Type	Digital Input
Description	Each PAC receives input from a discrete DPH ADDRESS signal line which is hardwired to the PAC from the DPH box located in the adjacent Cab. This ADDRESS line is used to determine if the DPH device has been activated by the driver.

12)PEI ADDRESS Lines

Signal Name	DPH ADDRESS Lines
Type	Digital Input
Description	Each PAC receives input from a discrete PEI ADDRESS signal line which is hardwired to the PAC from each of the three PEI boxes located within each Car. These ADDRESS lines are used to determine if the PEI devices have been activated by a Passenger pressing the PEI CALL button.

13)ANSM

Signal Name	ANSM
Type	Analog Input
Description	This interface is included to provide for the ability to measure the noise level within a car and automatically adjust the volume level as appropriate to maintain a consistent level of audio intelligibility and quality within a Car. To measure the car's ambient noise the PA sub-system is equipped with two automatic noise sensing (ANS) microphones, one connected to each PAC. The ANS Microphone monitors the ambient noise level within the car and provides an analog DCV output to the input of the PAC. The PAC will then sample the analog input at approximately 2 second intervals and calculate an ongoing weighted average of the most recent samplings. The sampling will be suspended during any PA announcements to alleviate errors in sampling. This calculated DC voltage value will then be compared to pre-set "MINIMUM" and "MAXIMUM" ANS Gain parameters, (The ANS Gain parameters are settable via the PTU). Based upon the results of this comparison, the appropriate ANS Gain level will be selected. This value will be processed through an analog to digital converter and the resulting value will be stored as a parameter in the PAC parameter table. *** Reference: At the time when an AVL or PA audio announcement is initiated, a separate software function of the PAC will utilize the current ANSM parameter value to adjust the audio output level of the volume control circuitry utilizing the I^2C interface. To guard against Noise created by the switching of mechanical relays, the PAC will mute the amplifiers during the switching of the volume control relays..

14)VOLUME CONTROL Signal

Signal Name	VOLUME CONTROL Signal
Type	Serial Output I2C
Description	The PAC volume control mechanism consists of both hardware and software components to manage different aspects of the functionality. With reference to the software functionality, the PAC generates a VOLUME CONTROL output signal which is used to deliver parameters and control codes to the Interior, Left Exterior, and Right Exterior volume control units. This function monitors all parameters set by the various Communication System devices, assembles a data packet reflecting the appropriate control codes

	necessary to invoke the desired functionality, and sends the data over the I2C serial interface to the Volume Control Units.

15)PAC_ID ADDRESS

Signal Name	PAC_ID ADDRESS
Type	Digital Input
Description	This identifies each PAC as being located in either CRA. The pin is hardwired in the chassis to be “0” at PAC-MA and “1” at PAC-MB.

16)USB_MP3

Signal Name	USB_MP3
Type	Serial Data Input and Output
Description	This is a USB port that connects the PAC’s on-board MP3 Player memory directly to the CNCS. It is used to transfer Audio Content files from the CNCS HDD to the local PAC’s MP3 Audio Player memory module.

3.1.1 DPO

1) Internal

Signal Name	Internal
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via interior speaker when signal which informs internal button has been pressed enters. The interior manual announcement is only available when the button has been pressed.

2) External

Signal Name	External
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via exterior speaker when signal which informs external button has been pressed enters. The exterior manual announcement is only available when the button has been pressed.

3) BOTH

Signal Name	Both
Type	Digital Input

Description	Generates alarm tone which informs manual announcement is in progress via interior and exterior speaker when signal which informs both buttons has been pressed enters. The interior and exterior manual announcement is only available when the button has been pressed.

4) IC

Signal Name	IC
Type	Digital Input
Description	LED of the IC button is usually turned off. If intercom communication is required and the button has been pressed, the LED of the IC button starts to blink, and recognizes that driver or crew requires intercom communication. Intercom communication will be made when pressing the blinking IC button. All requests and communication status will be cancel when cancel button has been pressed.

5) PEI

Signal Name	PEI
Type	Digital Input
Description	LED of the PEI button is usually turned off. If intercom communication is required and the button has been pressed, the PEI LED of the DPO starts to blink, and recognizes that driver or crew requires intercom communication. Intercom communication will be made when pressing the blinking PEI button. All requests and communication status will be cancel when cancel button has been pressed.

6) CANCEL

Signal Name	Cancel
Type	Digital Input
Description	Communication request with intercoms, communication status between intercoms, communication request of PEI, communication status with PEI, communication request with the OCC, and communication status with the OCC will be cancel when cancel button has been pressed.

7) OCC

Signal Name	OCC
Type	Digital Input
Description	LED of the OCC button is usually turned off. Requesting communication with the OCC is available by pressing the OCC button when DPO wants to communicate with the OCC or communication with passenger through PEI. LED of the OCC button blinks at this time. LED of the OCC button will be turned on if the communication with the OCC has been made. LED of the OCC button will blink when the OCC requests communication to the DPO. Every call including communication request with the OCC, communication

	status will be canceled when the cancel button has been pressed, and the LED of the OCC button will be turned off.

3.1.2 DPH

1) Internal

Signal Name	Internal
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via interior speaker when signal which informs internal button has been pressed enters. The interior manual announcement is only available when the button has been pressed.

2) External

Signal Name	External
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via exterior speaker when signal which informs external button has been pressed enters. The exterior manual announcement is only available when the button has been pressed.

3) BOTH

Signal Name	Both
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via interior and exterior speaker when signal which informs both buttons has been pressed enters. The interior and exterior manual announcement is only available when the button has been pressed.

3.1.3 **PEI**

Signal Name	PEI Call
Type	Digital Input
Description	Generates alarm tone which informs manual announcement is in progress via interior speaker when signal which informs internal button has been pressed enters. The interior manual announcement is only available when the button has been pressed.

3.1.4 CCP

1) CAB ON

Signal Name	CAB ON
Type	Digital Input
Description	Each CCP utilizes the adjacent Cab's CABON signal to determine if it should be set as the Active CCP for the entire Consist. The CAB ON signal will be derived from the Cab Make-Up Relay (CMR) by monitoring the output voltage of the CAB ON signal line. If the CAB ON signal is present, then the specified CCP will be designated as the Active CCP. As such it will provide the operational interface to the driver.

2) RS-485 Car Line

Signal Name	RS-485 CAR LINE
Type	Digital Input / Output
Description	The RS-485 Car Line data Input & Output signal is used to provide for digital communication between the PAC's and the various devices within a Car. The RS-485 Car Line provides for data communication between the Primary PAC, CCP, Display Signs, CNCS, LIC, and the secondary PAC in the same car. Within any given car the Primary PAC maintains protocol control of the RS-485 Car Line network. All other RS-485 devices within the Car will typically be polled within a continuous loop by the Primary PAC. In most cases the primary PAC will issue requests for specific responses from each device, and the addressed device will provide the requested response. In certain cases it will be necessary for the Primary PAC to temporarily pass RS-485control to another device so that it may affect the desired data communication. Additionally, the CCP will continually monitor the RS-485 port for relevant information which will be broadcast by other CS devices.

3) CCP GUI Keyboard Input

Signal Name	CCP GUI Interface Keyboard Input
Type	Set of logical values
Description	This is the primary operational interface by which Crew members select the desired functions, enter operational parameters, or review status data that is provided by the on-board communication system. By touching the screen directly over the displayed icons the Touch Screen records the specific screen location and equates the touch to represent a specific response. As such, the CCP touch screen is used by the Crew to select the desired function from the Menu Bar, or to enter specific data into fields that are presented by the various screens.

4) CCP ID Address Signal

Signal Name	CCP ID Address Signal
Type	Digital Input
Description	This identifies the CCP as being located adjacent to either CRA. The Address Line pin is hardwired in the harness to be “0” at PAC-MA and “1” at PAC-MB.

3.1.5 CNCS

1) RS-485 Car Line

Signal Name	RS-485 CAR LINE
Type	Digital Input / Output
Description	The CNCS is located in CRA of each Car. After the Inauguration Process is completed the CNCS will communicate with the other CS devices utilizing the RS-485 Car Line.

2) UDP communication is utilized to receive status data and to send and receive operating commands related to the sub-software components.

Signal Name	UDP Communication
Type	UDP data Input
Description	Utilize UDP protocol receive the status as well as to send and receive operating commands from each sub software component within the CNCS.

3) HDD

Interface Name	HDD
Type	File Input / Output
Description	The HDD located within the CNCS is the storage media used to store S/W components, schedule files, and various content files needed by the CNCS and other devices for ongoing operations. The CNCS software components utilize the HDD as the primary data source and destination relative to the storage and retrieval of data files. Files which are downloaded from the WLAN to the CNCS are stored to the HDD for later distribution. Data that is collected or compiled by the CNCS is stored into files on the HDD for later review or transfer to the WLAN.

4) LIC – RS232

Signal Name	LIC RS232
Type	Digital Input / Output
Description	This interface is used to provide for a direct data channel between the CNCS and the LonWorks Interface Card (LIC) for access to the MDS. This channel is used to deliver the MDS data from the MDS to the CNCS prior to sending the data to the Wayside via FTP. It is also used to transmit the CS software version and the GPS time to MDS.

5) Wireless LAN to Wayside

Signal Name	WLAN
Type	802.11g Wireless LAN connection
Description	Provides for wireless LAN connectivity between the onboard communications equipment and the Wayside network infrastructure.

6) USB link to PAC MP3 Memory

Signal Name	USB-MP3
Type	Standard USB 2.0 protocol
Description	Provides for a direct USB connection between the CNCS and the PAC's onboard USB Memory module. This allows for the direct copying of audio files for use by the MP3 audio player.

7) CNCS-LVDS Video Link

Signal Name	LVDS Video Signal
Type	ANSI/TIA/EIA-644 Low Voltage Differential Signaling (LVDS) Interface
Description	Video output signal from the CNCS to the VTX board

3.1.6 VTX

VTX H/W sets up its own ID through VTX_ID Address. Creates Protocol by the Protocol Generator Circuit and responds its own status with the PAC equipment via RS-485.

1) RS-485 Communication Port

Signal Name	RS-485 Serial Data Protocol
Type	Digital Input / Output
Description	Data Input / Output

2) VTX_ID Address

Signal Name	VTX_ID Signal
Type	Digital Input
Description	Scan the device connector pins to determine the exact location of the device within the vehicle. The unique VTX_ID is assigned based upon this location from a lookup table.

3.1.7 VRX

VRX H/W sets up its own ID through VRX_ID Address. Creates Protocol by the Protocol Generator Circuit and responds its own status with the PAC equipment via RS-485.

1) RS-485 Communication Port

Signal Name	RS-485 Serial Data Protocol
Type	Digital Input / Output
Description	Data Input / Output

2) VRX_ID Address

Signal Name	VRX_ID Signal
Type	Digital Input
Description	Scan the device connector pins to determine the exact location of the device within the vehicle. The unique VRX_ID is assigned based upon this location from a lookup table.

3.1.8 FDI

The FDI software component controls the actual LED display, and provides for the functionality allowing the device to display Route#, Destination information. The S/W also facilitates RS-485 communication with the PAC controller to provide for content updating, content management, and system diagnostics.

1) FDI RS-485 Communication Port

Signal Name	**FDI RS-485 CAR LINE**
Type	Digital Input / Output
Description	The RS-485 Car Line data Input & Output signal is used to provide for digital communication between the FDI devices and other RS-485 devices within a Car. Within any given car the Primary PAC maintains protocol control of the RS-485 Car Line network. All other RS-485 devices within the Car will typically be polled by the Primary PAC within a continuous loop. In most cases the primary PAC will issue requests for specific responses from each device, and the addressed device will provide the requested response. In certain cases it will be necessary for the Primary PAC to temporarily pass RS-485 control to another device so that it may affect the desired data communication.

2) FDI_ID Address

Signal Name	FDI_ID Signal
Type	Digital Input

Description	This signal is used to establish a unique FDI_ID parameter and as such to assign each FDI with a predetermined and specific RS-485 ID addresses. By scanning the connector pins, the FDI can determine the exact location within the Car where the specific FDI is installed. It will then establish an RS-485 Device _ID based upon parameter data retrieved from the local non-volatile memory.

3.1.9 SDI

The SDI software component controls the actual LED display, and provides for the functionality allowing the device to display Route#, Destination information. The S/W also facilitates RS-485 communication with the PAC controller to provide for content updating, content management, and system diagnostics.

1) SDI RS-485 Communication Port

Signal Name	SDI RS-485 CAR LINE
Type	Digital Input / Output
Description	The RS-485 Car Line data Input & Output signal is used to provide for digital communication between the SDI devices and other RS-485 devices within a Car. Within any given car the Primary PAC maintains protocol control of the RS-485 Car Line network. All other RS-485 devices within the Car will typically be polled by the Primary PAC within a continuous loop. In most cases the primary PAC will issue requests for specific responses from each device, and the addressed device will provide the requested response. In certain cases it will be necessary for the Primary PAC to temporarily pass RS-485 control to another device so that it may affect the desired data communication.

2) SDI_ID Address

Signal Name	SDI_ID Signal
Type	Digital Input
Description	This signal is used to establish a unique SDI_ID parameter and as such to assign each SDI with a predetermined and specific RS-485 ID addresses. By scanning the connector pins, the SDI can determine the exact location within the Car where the specific SDI is installed. It will then establish an RS-485 Device_ID based upon parameter data retrieved from the local non-volatile memory.

3.1.10 PII

This software is consistent with the FDI and SDI software but also monitors the AVL messaging system commands to display automated route, way finding, and emergency passenger information notifications.

1) PII RS-485 Communication Port

Signal Name	RS-485 CAR LINE
Type	Digital Input / Output
Description	The RS-485 Car Line data Input & Output signal is used to provide for digital communication between the PII and other RS-485 devices within a Car. Within any given car the Primary PAC maintains protocol control of the RS-485 Car Line network. All other RS-485 devices within the Car will typically be polled by the Primary PAC within a continuous loop. In most cases the primary PAC will issue requests for specific responses from each device, and the addressed device will provide the requested response. In certain cases it will be necessary for the Primary PAC to temporarily pass RS-485 control to another device so that it may affect the desired data communication.

2) PII_ID Address

Signal Name	PII_ID Signal
Type	Digital Input
Description	This signal is used to establish a unique PII_ID parameter and as such to assign each PII with a predetermined and specific RS-485 ID addresses. By scanning the connector pins, the PII can determine the exact location within the Car where the specific PII is installed. It will then establish an RS-485 Device _ID based upon parameter data retrieved from the local non-volatile memory.

3.2 Functional Requirements

3.2.1 PAC

When power is applied to the CS, each PAC will perform a series of initialization processes. The following diagram is intended to provide an overview of the PAC software components and the sequence of flow throughout the PAC initialization process.

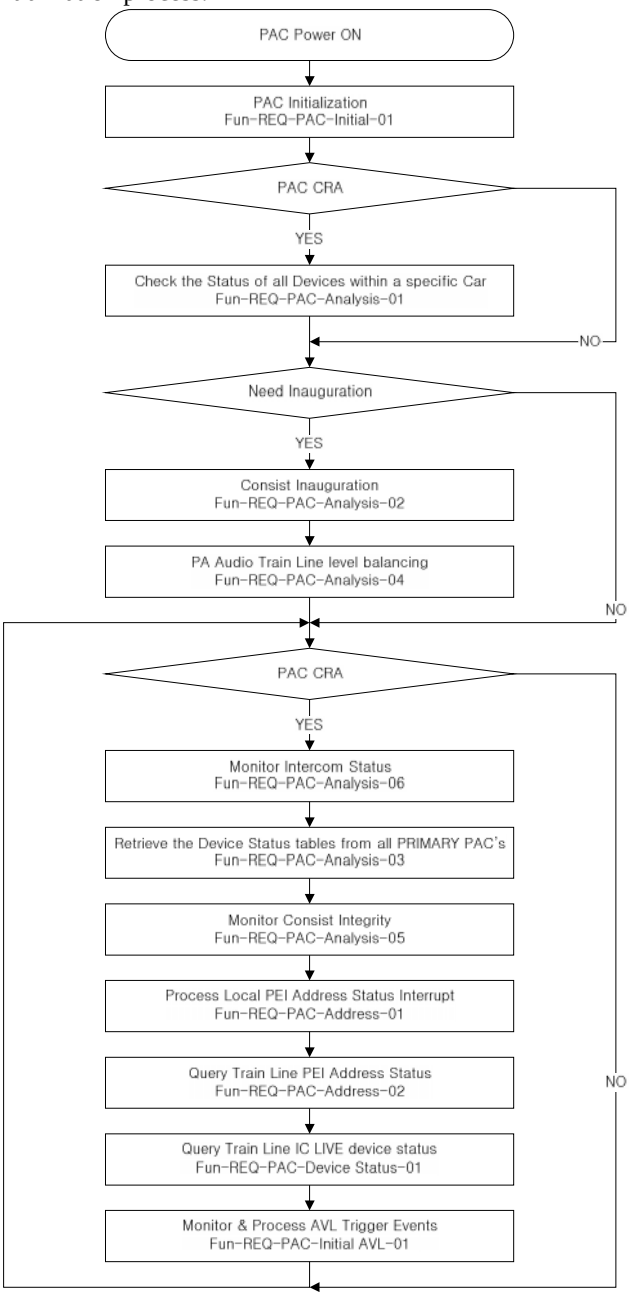


Diagram 3.1.1.2-A: Overview & Flow of PAC Functional Components

1) PAC Initialization

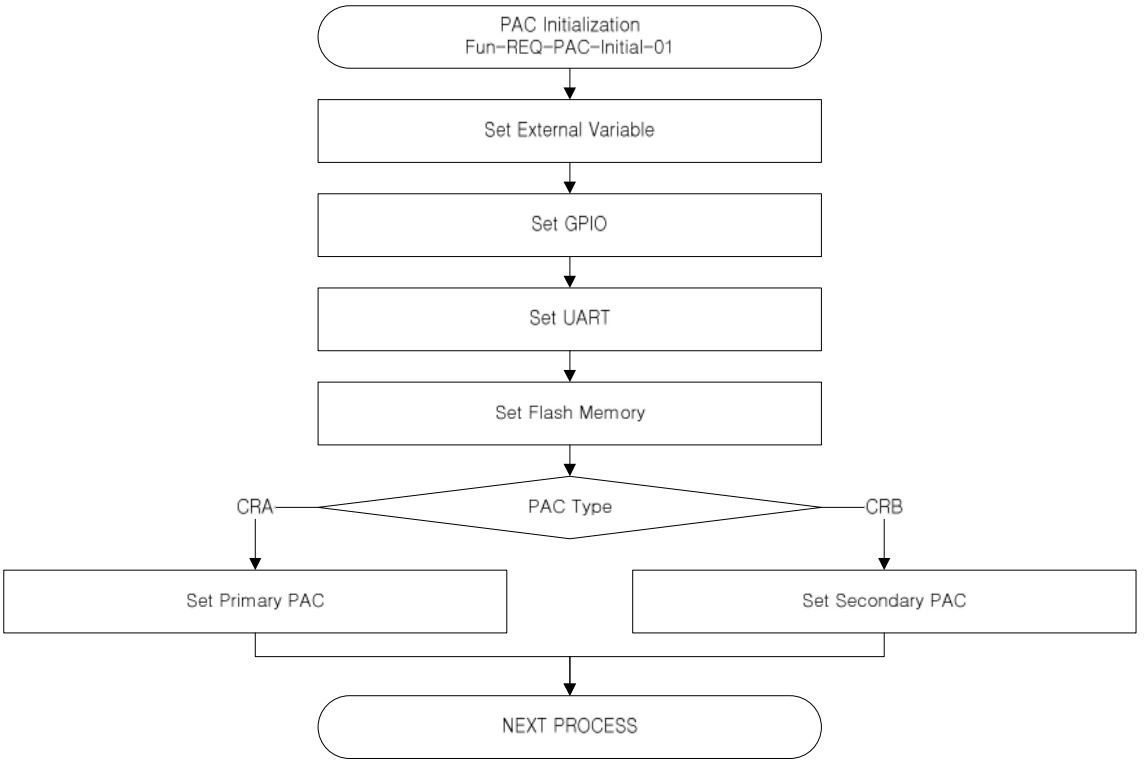


Diagram 3.1.1.2-B: PAC Initialization Flow Chart

ID	**Fun-REQ-PAC-Initial-01**
Function	Initializes all components & parameters inherent to the PAC device
Description	This functional requirement definition includes elements necessary to initialize a number of operating parameters, perform a set of default system tests, and place each PAC into a default operating mode. 1) Scan the PAC_ID Address line to determine if it is located in CRA. Set the PAC_ID Parameter. 2) All PACs identified to be PAC-MA must set an internal parameter indicating that they are to initially operate as the PRIMARY PAC. All PACs identified as PAC-MB must set an internal parameter indicating that they will initially operate in the SECONDARY PAC mode. The PRIMARY & SECONDARY designations are made with reference to the two PACs located within a Single and/or Married Pair Car. The PRIMARY designation indicates which PAC has priority within each Car and should not be confused with the designations of MASTER and SLAVE PACs. 3) Each PAC sets a separate parameter indicating that it will initially operate in the default SLAVE mode. MASTER and SLAVE designations are used to identify PAC priority within an entire Train Consist. All PACs will operate in the SLAVE mode with the exception of the one PAC within the Consist that is directly connected to the Master Cab, (The Cab in which the Driver has entered his key and placed it into the CAB ON state). 4) Evaluate and record the state of the COUPLER Signal as the COUPLER Parameter to indicate if the adjacent Coupler is in an open or closed position. If the Coupler Signal state changes, then the PAC must immediately update the COUPLER Parameter to indicate the correct state. 5) If the COUPLER parameter indicates an “Open” state, then set a TOC (Tail of Consist) Parameter indicating that the specific PAC has a TOC token (Flag). 6) Retrieve the RS-485 Car Line Address ID table from nonvolatile local memory. a) Attempt to establish RS-485 Car Line communication between the two sister PACs, (PAC-MA and PAC-MB). Perform a series of retries if initially unsuccessful in case the different PAC devices require varying lengths of time to complete their own POST processes. b) Create a Pass or Fail status parameter depending upon the results. c) Repeat this process for each of the remaining RS-485 Car Line devices. d) If any of the devices have successfully responded to the request for status command, then retry to establish communication with the Sister PAC. e) Report the Status data to the MDS 7) Retrieve the RS-485 Train Line Address ID table from nonvolatile local memory. a) Issue a command to OPEN the RS-485 Train Line relay on the Coupler side of the PAC leaving the line between the two PACs intact, (The Sister PAC will similarly be performing the same function). b) Attempt to establish RS-485 Train Line communication between the two Sister PACs, (PAC-MA and PAC-MB). c) Create a Pass or Fail status parameter depending upon the results. d) Report the Status data to the MDS 8) Check for the existence of a valid CAR NUMBER Parameter stored within its internal memory. If no valid CAR NUMBER parameter exists, then Step #9 must

	be performed, if a valid CAR NUMBER does exist then utilize the existing Parameter and skip to Step #9 below. 9) If the Car Number parameter doesn't exist, PAC-MA and PAC-MB will each receive the car number from the MDS in a different manner. a) PAC-MA will request and retrieve the Car Number from the MDS via the RS-485 Car Line connection to the LIC and thus the MDS. After the number has been received and stored as the CAR NUMBER parameter in local memory, the PAC-MA will send an RS-485 telegram to PAC-MB containing the appropriate number. b) PAC-MB relies on PAC-MA to provide the CAR NUMBER information. 10) Both PAC-MA and PAC-MB will perform an analysis of status data and test results to evaluate the viability of PAC-MA as the PRIMARY PAC. If PAC-MA is determined to be in a failed state, then PAC-MB will replace PAC-MA as the PRIMARY PAC and set its operating parameter accordingly. PAC-MA will either a) set itself as SECONDARY PAC if possible and appropriate, or b) be considered as disabled if its watchdog circuitry has shut down the RS-485 connectivity. 11) Evaluate the CAB ON Parameter. 12) Execute *Fun-REQ-PAC-Analysis-01*. This is the second phase of the PAC initialization process and performs an analysis of all devices within the specific vehicle.

2) Check the Status of all Devices within a specific Car

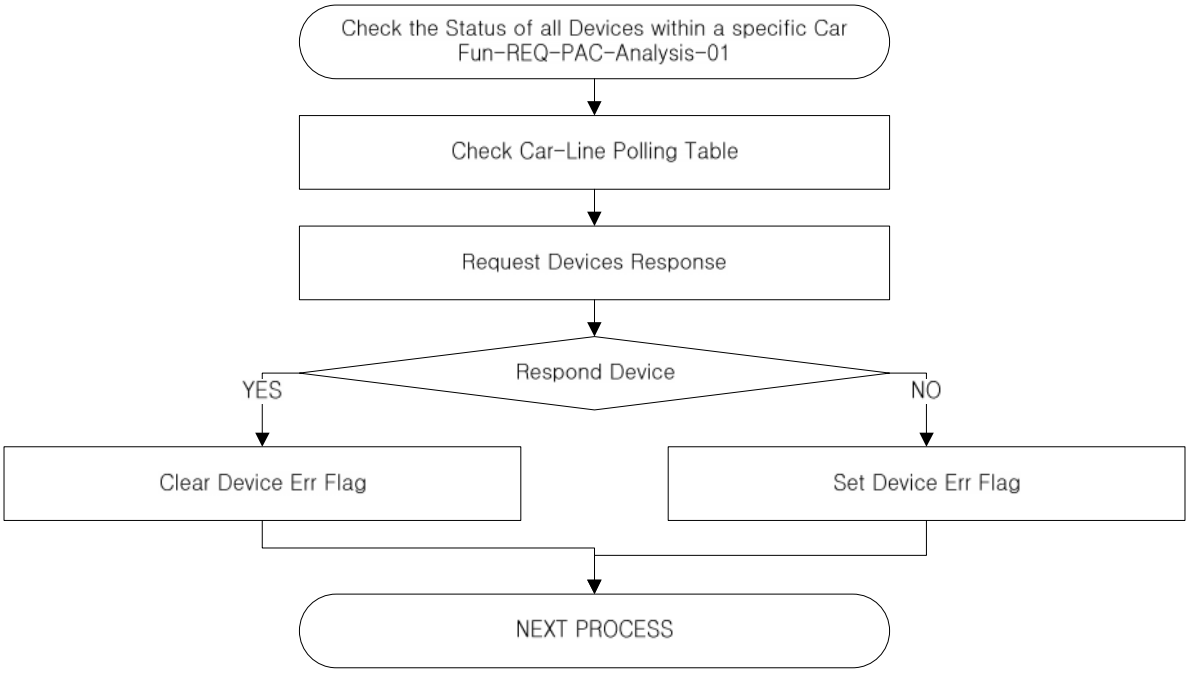


Diagram 3.1.1.2-C: RS-485 Car Line Device Analysis

ID	**Fun-REQ-PAC-Analysis-01**
Function	Poll and record the status of all RS-485 Car Line devices
Description	All Primary PACs perform the function of checking the Status of all devices within a specific Car. The PAC accomplishes this by utilizing the RS-485 Car Line infrastructure to poll each predefined device within the Car and attempts to retrieve its status data. 1) Load the RS-485 Car Line Address Table data & Polling Sequence data from nonvolatile memory. These tables will include specified polling loops, sequential device lists by polling loop, and each device's corresponding RS-485 Address ID. 2) Select the first specified loop & device. (Process sequentially 1 through N)

1										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	VTX	CNCS_M	CCP_M	VRC_M	SDI3	PIS_ALL
2										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	CCP_S	CNCS_M	CCP_M	VRC_M	SDI4	PIS_ALL
3										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	VRC_S	CNCS_M	CCP_M	VRC_M	PID11	PIS_ALL
4										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	PII1	CNCS_M	CCP_M	VRC_M	PID12	PIS_ALL
5										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	PII2	CNCS_M	CCP_M	VRC_M	VOIPC_S	PIS_ALL
6										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	FDI1	CNCS_M	CCP_M	VRC_M	PID13	PIS_ALL
7										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	FDI2	CNCS_M	CCP_M	VRC_M	PID14	PIS_ALL
8										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI1	CNCS_M	CCP_M	VRC_M	PID21	PIS_ALL
9										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI2	CNCS_M	CCP_M	VRC_M	PID22	PIS_ALL
10										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	PII3	CNCS_M	CCP_M	VRC_M	PID15	PIS_ALL
11										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	PII4	CNCS_M	CCP_M	VRC_M	PID16	PIS_ALL

12										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI5	CNCS_M	CCP_M	VRC_M	PID17	PIS_ALL
13										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI6	CNCS_M	CCP_M	VRC_M	PID18	PIS_ALL
14										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI7	CNCS_M	CCP_M	VRC_M	PID23	PIS_ALL
15										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	SDI8	CNCS_M	CCP_M	VRC_M	PID24	PIS_ALL
16										
CNCS_M	CCP_M	VOIPC_M	PAC_S	LIC_M	CNCS_S	CNCS_M	CCP_M	VRC_M	LIC_S	PIS_ALL

3) Select specified device address.
4) Request device status via RS-485 Car Line
5) Does device respond to request (Y/N)
   a) If YES: Record the retrieved Device Status in the local PAC status table.
   b) If NO: Retry request three times, if there is still no response then initiate the error handler and record the fault in the PAC status table
6) Select next Device and return to step #3 until all devices have been polled.
7) Select the next specified loop & device, and return to step #3 above. (Process sequentially 1 through N). When all loops are complete, STEP forward.
8) All device status for the specified Car will now be included within the Device Status Table stored in the local PAC's internal memory.

Number of devices per Car	Scan time per each status request	Number of Retries	Approximate cycle time per Car
Max 36	100ms	3	About 15 sec

9) Evaluate CAB ON Parameter (Skip this step if a Valid Consist Exists)
   a) If CAB ON = "1" (True): Launch *Fun-REQ-PAC-Analysis-02*
   b) If CAB ON = "0" (False): Launch *Fun-REQ-PAC-Analysis-04*
10) Repeat the scan cycle at step #2 based upon "Cycle timer" and "Interrupts"

3) Consist Inauguration

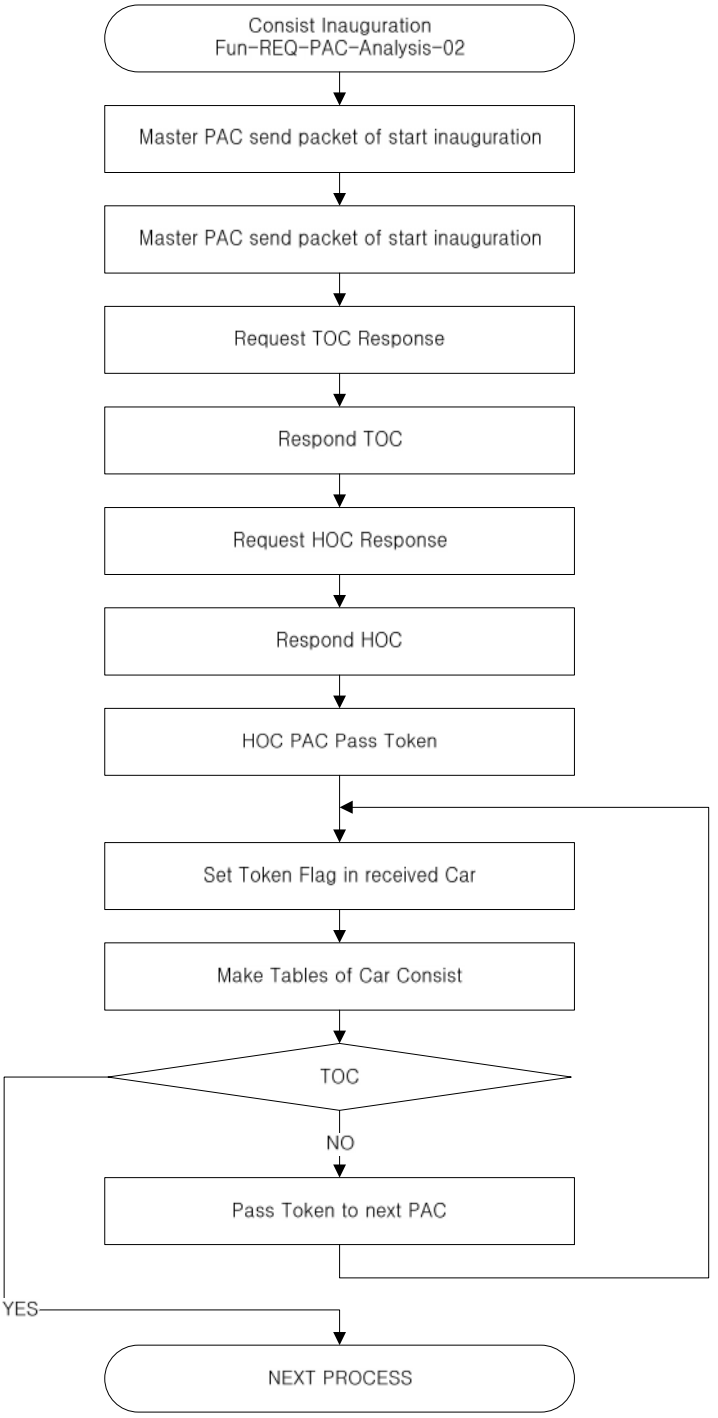


Diagram 3.1.1.2-D: Consist Inauguration Process Flow Chart

ID	**Fun-REQ-PAC-Analysis-02**
Function	Develop the Train Consist Configuration Table by utilizing the RS-485 Train Line infrastructure to discover and sequence all coupled Cars and attached PAC devices. Store the completed table on all PAC devices.
Description	This functional element performs the actual Train Consist Inauguration process and compiles a consolidated Consist Configuration Table. This process is performed by the Master PAC which writes the data table to its internal memory and replicates the table to all other PAC's in the Consist. Diagram 3.1.1.2-D above, (Consist Inauguration Process Flow Chart), illustrates the sequence of steps which are performed by the Master PAC throughout the Inauguration process. The following items represent a summary of the entire process. 1) Given that the CAB ON signal is present, set this PAC as the Master PAC. 2) Does the Master PAC hold a TOC Token, (Is the Coupler Open)? a) If YES, Create an HOC token on the Master PAC (It is Head of the Consist) b) If NO, Open the RS-485 Train Line relay toward the sister PAC, and issue a request for response from the TOC Token holder. The PAC that responds is the true HOC. Instruct the responding PAC to set itself as the HOC and record its PAC_ID and Car# in the Consist Configuration table. 3) Close all RS-485 relays and issue a request for response from the remaining TOC Token holder. Only one other Car in the Consist can possibly hold a TOC Token. Therefore, the PAC that responds is the true TOC. Record its PAC_ID and Car# into the Consist configuration table. 4) Place a PASS Token on the PAC holding the HOC token. 5) Pass the PASS token from the HOC to the TOC in a sequence of one PAC at a time. Record the PAC_ID, Car#, and Consist Position of each PAC before the Token is passed from one PAC to the next PAC. The RS-485 relay device state configurations and software processes necessary to accomplish this portion of the Inauguration are contained in the following table, (Table 3.1.1.2-A). 6) When the PASS Token and TOC Token reside on the same Car # and have the same PAC_ID, then the Token passing process is complete. 7) Finalize the Consist Configuration Table to the Local Master PAC memory. 8) Initiate function *Fun-REQ-PAC-Analysis-03* to consolidate all of the Device Status tables located on each of the PRIMARY PACs within the Consist back to the Master PAC for distribution to the CCP.

RS 485 Train Line Relay States During Inauguration

Inauguration Steps Input Signal	Married A 1nd		Married B 2nd		Married A 3nd		Married B 4nd		1	2	3	4	
	PAC A	PAC B	PAC A	PAC B	PAC A	PAC B	PAC A	PAC B					
	Ra	Rb	Ra	Rb	Ra	Rb	Ra	Rb					
1. Power ON	C	C	C	C	C	C	C	C	X	X	X	X	All Relays default to the close (on) State
2. Coupler Check	C	C	C	C	C	C	C	C	X	X	X	X	PAC - A of the 1st Car & PAC-B of 8th Car both hold a TOC token
3. Cab-ON @(2nd, PAC-A) SET Master PAC	O	C	C	C	C	C	C	C	X	X	X	X	PAC-A of the 2nd Car is the Master PAC.
4. Request TOC Response	O	C	C	C	C	O	C	C	1	X	X	4	Open(2nd,PAC-A,Rb)to disconnect from Paired PAC-B, Master PAC requests Toc response & (1st. PAC - A) responds.
5. Set HOC Token Flag	O	C	C	C	C	C	C	C	1	X	X	4	Master PAC sets (1st. PAC - A) HOC & then returns all relays to default status.
6. Request TOC Response	O	C	C	C	C	C	C	C	1	X	X	4	The only remaining remaining TOC token is held by (4th, PAC-B) & responds to the request
7. Get Response & Update Consist Inauguration Tables	O	C	C	C	C	C	C	C	1	X	X	4	All PAC's update their Inauguration table with HOC, TOC, and Master Info.
8. Pass Token (1ST. PAC-B)	O	C	C	O	O	C	C	O	1	X	X	4	The PAS Token is passed from the HOC token holder to the next adjacent PAC.
9. Get Response & Update Consist Inauguration Tables	O	C	C	C	C	C	C	C	1	2	X	4	All PAC's update their Inauguration table to include (2st. PAC-B) paramaters.
10. Pass Token (2nd, PAC-A)	O	C	C	C	C	O	O	C	1	2	X	4	The PAS token is passed from the current token holder to the next adjacent PAC.
11. Get Response & Update Consist Inauguration Tables	O	C	C	C	C	C	C	C	1	2	3	4	All PAC's update their Inauguration table to include (3st. PAC-B) paramaters. Inauguration complete.

C = Closed	TOC Token Flag		Master PAC
O = Open	HOC Token Flag		Update Consist Table
	Passing Token		Temp TOC Token

Table 3.1.1.2-A: Consist Inauguration, RS-485 relay State table

4) Retrieve the Device Status tables from all PRIMARY PAC's.

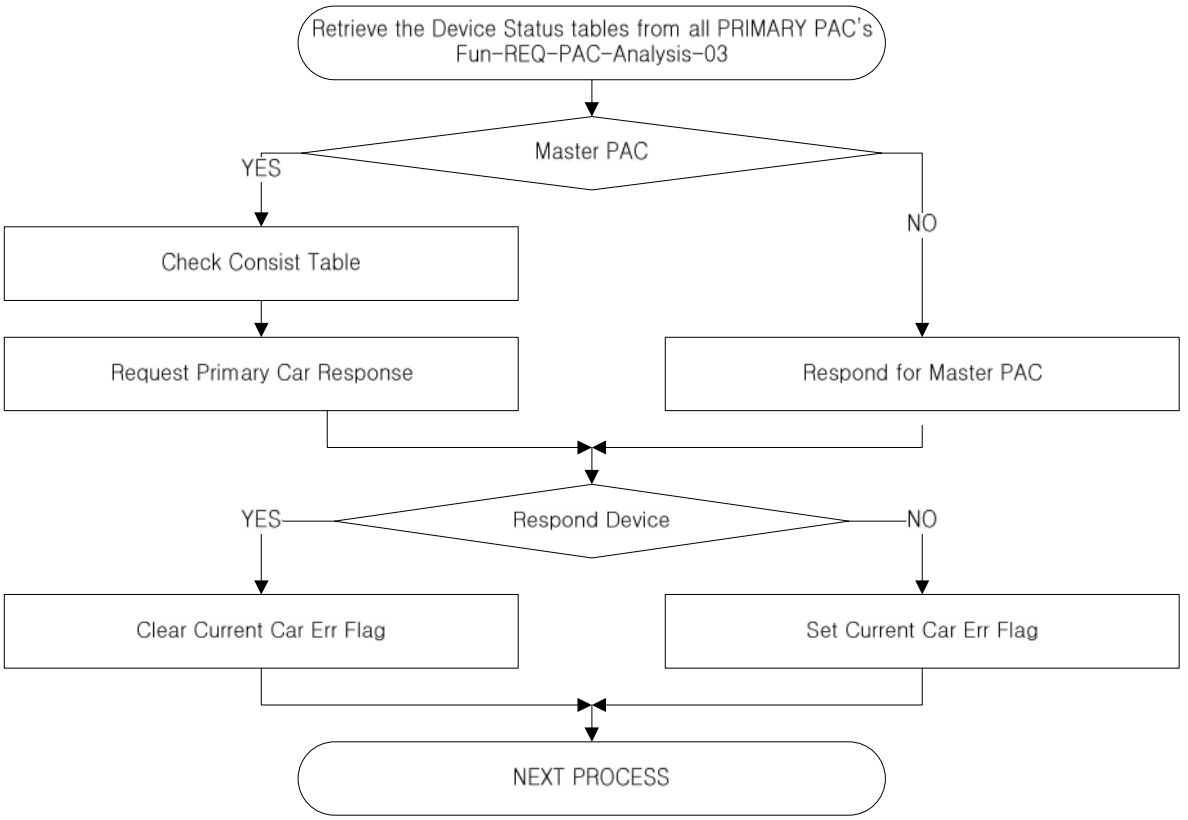


Diagram 3.1.1.2-E: Master PAC – Consist PAC Device Status Table retrieval

ID	Fun-REQ-PAC-Analysis-03
Function	Retrieve the Device Status table from the PRIMARY PAC of each Car in the Consist and record the data in the Master PAC's Device Status table.
Description	Upon the successful completion of the Inauguration Process, the Master PAC will have a valid Consist with confirmed RS-485 Car Line continuity. As such, it can perform the function of retrieving the Device Status tables from each of the PRIMARY PACs throughout the Consist. The Master PAC must poll each of the PRIMARY PACs listed in the Consist Configuration table, and then aggregates the data within the Master PAC. Once aggregated in the Master PAC all Consist Device Status data is routed to the CCP where it can be viewed by the Driver. 1) Retrieve the appropriate data from the Consist Configuration Table to develop a sequential list of all PRIMARY PAC devices within the Consist. The list will identify Cars 1 through N of the Consist, including each Car's position in the Consist, its true Car #, and the device ID of each PRIMARY PAC. 2) Select the first PAC_ID in the sequential list, and retrieve its RS-485 ID. 3) Is the selected PRIMARY PAC also the MASTER PAC? a) YES: Copy the Local Car's device status table data to the Consist device status table. b) NO: The Master PAC sends a "Request to Send the Device Status Table" to the specified PAC. The specified PAC responds by sending the requested data. The Master PAC appends the retrieved data to the Consist device status table. 4) Process any errors via an error handler 5) Select the next PAC_ID in the sequential list, retrieve its RS-485 ID, repeat step #3 until all remaining PRIMARY PAC's in the Consist have been polled. 6) Select the RS-485 Device ID of the CCP 7) Send the consolidate Consist Device Status Table data to the CCP for display and to the CNCS to be written in a Log file.

5) PA Audio Train Line level balancing

ID	Fun-REQ-PAC-Analysis-04
Function	The Master PAC initiates a process that injects a known "Reference Tone" onto the PA audio train line which is then used by each PAC in the Consist to establish a "PA Train Line Gain" parameter. This will ensure a consistent train line audio level.
Description	This component is provided to meet the technical specification requiring that the PA audio level of the Train Line not sustain a loss in excess of 3db across the entire Consist. To accommodate this requirement, the PAC design includes a signal generator which is capable of producing a 11KHz "Reference Tone". The Master PAC will inject this "Reference Tone" onto the PA Audio train line for use by all SLAVE PACs as a known reference level. Each SLAVE PACs will then discover the

	actual output level of the PA audio line and compare it to the parameter defined "PA Train Line Output Level", and calculate an appropriate "PA Train Line Gain" parameter. This element must function in two different ways depending upon whether a specific PAC is operating in the MASTER or SLAVE mode. •<u>MASTER PAC:</u> 1) Utilize the onboard tone generator to create a 11KHz tone. 2) Feed the 11KHz signal onto the PA Audio train line at an output level which is consistent with a pre-defined parameter for the PA TESTsignal. 3) Sustain the Test Tone for a sufficient length of time to allow each of the other PACs in the Consist to detect the 11KHz tone, evaluate the ensuing output level, and establish an appropriate "PA Train Line Gain" parameter. •<u>SLAVE PAC</u>: 1) Load the "PA Train Line Output Level" parameter from the PAC memory. 2) Monitor the PA Audio train line for a 11KHztone. Once the 11KHz tone is detected for a specified period of time, then measure and record the audio line level as a reference to evaluate the sustained db loss at that point. 3) Compare the current output level at each PAC to the level specified in the "PA Train Line Output Level" parameter. 4) The MASTER PAC will establish the overall PA Audio line loss across the Consist by requesting the TOC PAC to report its level variance in comparison to the predefined reference level parameter value. 5) Is the overall loss across the Consist greater than 5db? a) *NO*: No level adjustment is required. Set the PA Gain parameter to = "0" b) *YES*: Calculate an appropriate audio level gain adjustment that may be applied by each PAC to maintain the line level within 3db across the Consist. 6) Establish a "PA Train Line Gain" parameter to reflect the determined gain adjustment, and store it in the local memory of each PAC. 7) The PACs will apply this adjustment level for all PA announcements. **Note: The specific algorithm utilized to calculate an appropriate gain level will be determined by the hardware engineer during the development process.

6) Monitor Consist Integrity

ID	Fun-REQ-PAC-Analysis-05
Function	Monitor Consist Integrity, Coupler Status and CAB ON Signal
Description	This function continually evaluates the integrity of the Consist by monitoring the current states of the CAB ON and COUPLER signals. Any state change relative to either signal will require some form of analysis and/or resulting action. In the event that a CAB ON signal state change from “1” to “0” is detected, then the MASTER PAC must set a timer to evaluate the length of time the “0” state persists. If the time interval is determined to be within a specified parameter value range, then the existing Consist Configuration shall be considered to remain valid. If the time interval exceeds a parameter defined limit, then the PAC will re-inaugurate. In the event that a COUPLER state change is detected by any PAC in the Consist, then a similar analysis must be performed. Relatively short COUPLER state changes, (parameter defined duration), may be considered to be inconsequential, while longer ones must result in further analysis, notifications, error logging, and re-inauguration.

7) Monitor Intercom Status

ID	Fun-REQ-PAC-Analysis-06
Function	Evaluate the IC LIVE Lines signal states to provide device Status Data
Description	This function monitors the 12 hardwired IC LIVE signal lines which are direct inputs to the PAC from the two DPO’s, two DPH’s, and eightPEIcontrol boxes. Each of these IC LIVE signal lines must be scanned and evaluated on an ongoing basis to indicate if the device is functional on or not. If a DCV signal is present, then the device is powered on and is ready for operation. This signal does not indicate that an IC Call is in progress, but merely indicates that the device is powered on and is operational. The value of each IC LIVE Line signal state corresponding to the 12IC devices must be stored as separate parameters within the in-memory PAC Parameter Table. The parameter table values must be regularly updated to reflect the state of the corresponding physical device’s IC LIVE Line. *Note: “0” and “1”: (“1”= DPO/ DPH/PEI box is Operational, “0” = Non-Operational)

8) Process Local PEI Address Status Interrupt

ID	**Fun-REQ-PAC-Address-01**
Function	Process Local PEI ADDRESS Status Interrupt
Description	When a PEI CALL button is pressed, the specific device will generate a 15VDCsignal on the PEI Address line. This function monitors the several PEI ADDRESS lines and updates parameters indicating the state of each Address line. It writes a value representing the device state to the PAC Parameter table located within the local memory of the PAC. Values: "0" = Inactive (No CALL is in progress on this specific PEI Box). "1" = Active (The CALL button has been pushed and a call is in progress)

9) Query Train Line PEI Address Status

ID	**Fun-REQ-PAC-Address-02**
Function	Query Train Line PEI Status
Description	1) The MASTER PAC retrieves the Local PEI ADDRESS status data elements from each primary PAC in the Consist and writes the data to the Master PAC Status Table. 2) The stored status data identifies all PEI devices in the Consist, indicating the physical car#, specific location within, and the PEI ADDRESS status of each device. 3) Compares the retrieved status values to the previously stored values to determine if any state changes have occurred. 4) In the event that any state change has occurred, sends a telegram to the CCP via the RS-485 Car Line indicating the state change event. 5) Queues the corresponding MP3 Audio File ID(s) to be played via the Automatic Broadcast player. 6) Forwards the appropriate status data to the CCP enabling the CCP to display the current device states and proper indications to the crew.

10) Query Train Line IC LIVE device status, (DPO/ DPH/PEI).

ID	**Fun-REQ-PAC-Device Status-01**
Function	Query IC LIVE Device States
Description	1) The MASTER PAC retrieves the Local IC LIVE status data elements from each primary PAC in the Consist and writes the data to the Master PAC Status Table. 2) The stored status data identifies all IC devices in the Consist, indicating the physical car#, specific location within, and the IC LIVE status of each device. 3) Compares the retrieved status values to the previously stored values to determine if any state changes have occurred. 4) In the event that any state change has occurred, sends a telegram to the CCP via the RS-485 Car Line indicating the state change event. 5) Forwards the appropriate status data to the CCP enabling the CCP to display the current device states and proper indications to the crew.

11) Monitor & Process AVL Trigger Events

ID	**Fun-REQ-PAC-Initial AVL-01**
Function	Monitor & Process AVL Trigger Events
Description	This function processes RS-485 Car Line data packets related to the handling of AVL messaging triggers. AVL triggers may reference any or all of the following types of massages and content: 1) Audio Announcements – Route Service Announcements 2) Audio Announcements – PEI Emergency Assistance Announcements 3) LED Sign Messages – Route Service Notifications 4) LCD Display – Route Service Notifications 5) LCD Display – Advertisement & Public Service Notifications As part of the AVL route processing performed by the Master CCP, the CCP generates an AVL trigger whenever the vehicles GPS coordinates are within a margin of error proximity of a defined trigger point coordinate pair. All relevant trigger information is formatted into a data packet and distributed via the RS-485 Car Line. The Master PAC must perform the following functions. 1) Evaluate all RS-485 data packets to determine if they represent a triggered event. 2) If so, determine the nature of the triggered event and initiate the appropriate AVL messaging action(s). 3) IF an audio message is indicated by the trigger, then load the PA system operating parameters associated with the trigger definition and the current system configuration. This may include speaker selection, volume level, muting control, and Gain control parameters. 4) IF this is the MASTER PAC: a) Serve as a gateway to route all trigger data packets from the RS-485 Car Line to the RS-485 Train Line. b) Pass initiation parameters to the MP3 Broadcast Player for any audio message(s) specified by the trigger. c) Route the output, (If any), of the MP3 player to the specified local amplifiers and to the PA Audio Train Line. 5) IF this is a SLAVE PAC: a) Serve as a gateway to route all trigger data packets from the RS-485 Train Line to the local RS-485 Car Line. b) If an audio signal has been placed onto the Audio Train Line by the MASTER PAC, then turn on the amplifiers as specified by the parameters and route the Train Line audio signal to the local amplifiers. 6) Monitor the door control state as required in order to confirm AVL trigger point authenticity decisions. 7) Adjust the volume of the interior / exterior speakers.

3.2.2 CCP

1) CCP Initialization

ID	**Fun-REQ-CCP-Initial-01**
Function	CCP Device initialization
Description	When the MASCON Key is placed in any of the "ON" positions, all CCP devices in all cars of the Consist will be supplied with power and begin their initialization process. The individual CCPs will respond differently depending on the following conditions. 1) Is the CCP directly connected to the MASTER PAC. If so, then it is designated as the MASTER CCP and will be used as the primary interface between the driver and the CS. 2) Is the CCP located in the same Car as the Master PAC but not directly connected. If so, it is designated as the SECONDARY CCP and could possibly be used to interface with the CS in a backup role. 3) The CCP is located in a different Car from the MASTER PAC. If so the CCP has no direct RS-485 Car Line connection to the MASTER PAC and as such can only support limited functionality. These will be referred to as PASSIVE PACs. Upon power up each CCP will perform a POST process followed immediately by loading the Windows CE operating system from its local memory module. Once the OS has been loaded the CCP will automatically execute the following sequence: I. Load the RS-485 Drivers II. Load the "CCP GUI" application software component. III. Evaluate the connector Signal to determine if the specific CCP is installed adjacent to CRA ~~or CRB~~ and set a CCP_ID parameter accordingly. IV. Set a parameter indicating the appropriate RS-485 Car Line ID V. Set a parameter indicating that the current Train Number = " " VI. Set a parameter indicating that the current Crew ID Number = " " VII. Restore all available operating parameters from the local Nonvolatile RAM. This will include the parameter indicating the physical Car number. If no Car number exists, then request it from the PAC via the RS-485. VIII. Evaluate the CAB ON signal to determine the appropriate operating mode: a) IF the CAB ON signal is present, then the CCP sets a parameter indicating that it will operate as the MASTER PAC. b) If the CAB ON Signal is not present then the CCP must further evaluate whether it is located in the same Car as the MASTER PAC and CCP. If "YES" then it will operate in a SECONDARY mode and will restore and evaluate any current operating parameters from memory and display the current operating parameters settings. If "NO" then it will operate in a PASSIVE mode and will only be able to display parameters relating to the specific Car in which it is located. The screen will remain dark unless touched. Once activated it will return to dark after

	75seconds of no activity. IX. Is this CCP the MASTER CCP: a) NO: Go To VIII b) YES: Execute the *Fun-REQ-CCP-Control-01* which will prompt for and set the train #. X. Idle Wait

2) Set Train Number

ID	**Fun-REQ-CCP-Control-01**
Function	Set Train Number
Description	This function provides for a mechanism by which the operator can enter the desired train number into the CCP GUI. Once entered, the number is validated against the master DENVER RTD train schedule and causes the CCP to retrieve the corresponding schedule elements and AVL trigger data from the CNCS and load it into the CCP memory. Upon initiation by touching the "TRAIN" button on the selection bar of the CCP GUI, it will perform the following functions: 1) The CCP must display the "Set Train Number" screen. 2) Prompt the operator to "Please enter Train Number" and simultaneously display a graphical key pad entry screen. 3) The driver then enters the train number via the presented Key pad and touches the "Enter" button to confirm the entry. The train number must be visibly displayed at the time of input and the operator must be provided with a "Clear" key to correct any erroneous entries. 4) Once the entry has been confirmed, the CCP must validate the entered "Train Number" against the current DENVER RTD operating train schedule. If the entry is not valid, then the operator must be notified of the error and prompted to re-enter a different Train Number. 5) If the entry is a valid "Train Number", then the CCP must retrieve the corresponding schedule elements and AVL trigger data from the CNCS and load it into the CCP memory.

3) Set Crew ID

ID	Fun-REQ-CCP-Control-02
Function	Set Crew ID
Description	This function provides for a mechanism by which the operator can enter his assigned crew ID into the CCP GUI. Once entered, the crew ID data is written as a parameter to the CCP memory and is also forwarded via the RS-485 Car Line to the CNCS for recording into the activity log file. This function is initiated by touching the "CREW" button on the selection bar of the CCP GUI. At the time of train inauguration, this function "Set Crew ID" is automatically invoked by the CCP initialization procedure right after a valid Train Number is entered by the operator. 1) The CCP must display the "Enter Crew ID" screen. 2) The operator will be prompted to "Please enter Train Number" and will simultaneously be presented with a graphical key pad entry application. 3) The driver enters the crew ID via the presented Key pad and touches the "Enter" button to confirm the entry. The crew ID must visibly display at the time of input and the operator must be provided with a "Clear" key to correct any erroneous entries. 4) Once the operator touches the "Enter" keypad to confirm the crew ID, the CCP will validate the entered "Crew ID". The CCP will display an error message to the operator to re-enter a valid crew ID if the entered crew ID is not valid. 5) Once a valid crew ID is entered, the Crew ID# is stored in the CCP's parameter table within internal memory, and forwarded to the CNCS for inclusion in the activity log file.

4) Set Route Information

ID	Fun-REQ-CCP-Control-03
Function	Set Route Information
Description	This function provides a mechanism by which the operator can initially view and then edit the route information on the CCP screen as it pertains to the current Train Number. The CCP will display the departure station stop, scheduled time of departure and any delays related to the schedule adherence for each station stop of the train route. This function is initiated by touching the "ROUTE" button on the selection bar of the CCP GUI. To change or skip station stops of the scheduled train route, the operator needs to touch the "ROUTE" button again. The CCP will then provide a different route screen whereby the operator can select: 1) "STOP": to stop at a specified train station of the train route. 2) "SKIP" : to skip one or more station(s) of the train route; 3) "EXP" : to specify an Express schedule without stopping at most stations of the scheduled train route. 4) "LOCAL": to change an Express train schedule to stop any every station. After each route change, the CCP will relay any relevant route change data via the RS-485 back to the PAC and CNCS. All changes are logged to the CNCS activity log file.

5) Set Configuration Parameters

ID	Fun-REQ-CCP-Control-04
Function	Set Configuration Parameters
Description	This function provides a mechanism by which the operator can change the PA audio volume level, the LCD panel's brightness, or reset the CS parameters to their initial default values. The function is initiated by touching the "CONFIG" button on the selection bar of the CCP GUI. This is the main "CONFIG" screen by which the user will be presented with various options for adjusting the system parameter settings. Three spare buttons are available for future features. The default level values pertaining to the NORMAL and SOFT parameter settings for both the interior and exterior speakers are set by a separate function via the PTU. This function merely allows the operator to select the NORMAL, SOFT, or MUTE operating modes. 1) To change the Speaker volume, the operator will select the "SPEAKER VOLUME" button on the "CONFIG" screen. The GUI will then display a separate screen allowing the operator to independently set the volume level of both the Interior and Exterior speakers across the Consist. The parameter settings must include the following choices: a) Interior Speakers: NORMAL SOFT b) Exterior Speakers: NORMAL SOFT MUTE 2) To change the LCD panel brightness, the operator needs to first touch the "CONFIG" button on the selection bar of the CCP GUI. Next, touch the "LCD DIMMER" button and a level adjustment screen will be displayed allowing the operator to chose the desired level via a sliding value bar within a range from 20% to 100%. 3) By selecting the "CONFIG" button followed by the "RESET" button, the CCP will restore the CS parameters to their initial default values. Upon completion, press the "HOME" button to return to the normal CCP operational review screen. The changed parameters will be stored to the CCP memory and sent via the RS-485 to the PAC for distribution throughout the Consist.

6) Display the status of RS-485 monitored devices throughout the Consist

ID	Fun-REQ-CCP-Control-05
Function	Display the status of RS-485 monitored CS devices throughout the Consist
Description	This function enables the driver or a crew member to quickly assess the status of the CS devices within the entire Train Consist on a Car by Car basis. By touching the "CHECK" button on the selection bar of the CCP GUI, a graphical representation of the current Consist and the status of the various sub-systems are displayed. Each car within the Consist is graphically depicted on a single screen allowing the operator to select a specific Car by touching the numbered Icon. The specific Car is then graphically depicted by illustrating the CS components within that Car as well as the current operating status of each device. Different colors and text descriptions are used to indicate different system fault conditions. Properly functioning components are labeled and depicted in green. Non-functional components are labeled and depicted in red. The exact color scheme to depict the different system faults will be defined in the SDDS. Exact message verbiage of the different system faults messages will be defined in the SDDS. The actual device states displayed are derived from device status data elements provided to the CCP by the MASTER PAC via the RS-485 Car line. The MASTER PAC maintains a parameter table which includes the current status of all monitored CS devices throughout the Consist.

7) Display the CCP Help Screens

ID	Fun-REQ-CCP-Control-06
Function	Display the CCP Help Screens
Description	This function is a mechanism by which the operator can briefly view the on-board quick reference instruction guide of the Denver RTD Eagle P3 vehicle. To start this function, the driver touches the "HELP" button on the selection bar of the CCP GUI. The HELP menu system is consist of seven sections. The sections are named "TRAIN", CREW", "ROUTE", "CONFIG", "CHECK", "HELP" and "PEI". Touch and select any of the seven HELP topics presented within the HELP menu screen. If the operator touches the "TRAIN" button on the HELP menu screen, a new HELP subtopic screen should appear that describes and guides the operator through entering the Train Number into the CCP GUI. Likewise if the operator touches and selects the "CHECK" button of the HELP menu screen, a new HELP subtopic screen should appear that describe and guide the operator in understanding the different color coding of all the components on the train.

8) Emergency PEI Call Notification

ID	Fun-REQ-CCP-Control-07
Function	Emergency PEI Call Notification
Description	This function provides visual notification to the driver or the operator in the event that a passenger presses a PEI Call button from any car of the Consist. When a passenger presses a PEI "CALL" button, the Master PAC will immediately be forwarded status data indicating both the car # and the specific PEI device # which has initiated the call. The PAC forwards this status data to the CCP which is converted into a visual message displayed by the CCP informing the operator that a passenger needs assistance in a specific Car number.

9) Monitors, Reads and Transmits RS-485 Car Line data

ID	Fun-REQ-CCP-Control-08
Function	Receives & Transmits RS-485-data
Description	This function provides for the ability to monitor and respond to RS-485 requests. The CCP must be capable of responding to normal RS-485 requests, and must be capable of initiating a request by setting a status flag indicating that it has information to send. The Primary PAC will respond by requesting that the CCP provide the required data according to the appropriate protocol. The following requirements and process elements must be handled by this function: 1) Load / adhere to RS-485 protocol definitions 2) Set Status and Function flags as necessary 3) Support RS-485 communication in either a "Normal" mode or in a "Bulk Data Transfer" mode. 4) Respond to requests for status initiated by either the Primary PAC or the CNCS. 5) Evaluate RS-485 data received and compare it to the protocol definitions to determine the nature of the data received and the action required. 6) Evaluate the specific command sequences specified 7) Assimilate the appropriate controls, protocols, and processes necessary to affect the desired actions. 8) Send RS-485 data responses as required and in the specified manner.

10) Set Status Update Flag

ID	Fun-REQ-CCP-Control-09
Function	Set Status Update Flag
Description	This function enables the CCP to set or clear a flag indicating that it either needs data from another source or has data to deliver to another source. Additionally this function must also incorporate the flexibility to set the operating mode of the CCP's RS-485 communication to either the "NORMAL" or "BULK DATA" mode.

11) Receive bulk data Mode

ID	Fun-REQ-CCP-Control-10
Function	Receive bulk data Mode
Description	The purpose of this function is to provide for a mechanism by which the CCP can be switched between operating in a “NORMAL” handshake mode or operating in a “BULK DATA” broadcast mode.

12) Display Current Route and Schedule Adherence data

ID	Fun-REQ-CCP-Display-01
Function	Displays Current Train Operational Information
Description	This function displays the normal schedule route of a train schedule. It allows the driver to review and edit the Train route in the event of unknown weather created or road closure or station stops conditions that may require schedule changes and other 1) The current Train number and corresponding Route number is displayed on the CCP’s LCD based on information retrieved from the train service schedule corresponding to the train number parameter input by the crew. 2) Station names corresponding to the designations of “Departure”, “Actual” and “Next1” as indicated in the train service schedule are displayed. Adjacent to each display entry, the function will also display the scheduled departure times, actual departure times, and delayed times standardized on the basis of present time.

13) AVL GPS Route Processor

ID	**Fun-REQ-CCP-Comm-02**
Function	Plot real time GPS coordinates against the prerecorded route coordinates.
Description	This function is used to track a train's progress through the specific scheduled route that has been selected by entering the Train Number into the CP. The CCP will compare real time Train location coordinates provided by the GPS against prerecorded route coordinate data as a means to plot the vehicles current position within the route. Each time a coordinate has been acquired, the CCP will compare that coordinate to the scheduled trigger point coordinates to determine if the acquired point is also a defined trigger point. 1) Retrieve the Train # from the CCP parameter table. 2) Extract the route definition coordinate data list, and the trigger point coordinate list from the master schedule data base file which corresponds to the selected route. 3) Compare the current location data to the route definition list data to identify the data element that represents the closest proximity to the current location and calculate the actual distance between the two points. 4) In the event that the measured distance to any given point is deemed to be less than a predefined margin of error distance from that given point, then the CCP will deem the specific point to have been acquired. 5) Each acquired pointed will be recorded and stored as the "Current Location" parameter to the CCP memory. This parameter will be updated to a new location value each time a subsequent coordinate location within the defined route has been acquired. 6) Compare the "Current Location" parameter to the "Trigger Point Coordinate List" to determine if the current location is a defined trigger point. 7) If the current location is a defined trigger point, then set a status flag indicating that a new trigger point has been acquired and indicate the specific trigger point number in a "Current Trigger Point" parameter.

14) AVL Event Trigger Processor

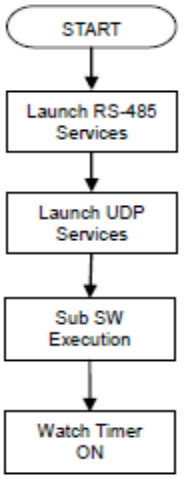
ID	Fun-REQ-CCP-Comm-03
Function	Continually evaluates the "Current Trigger Point" parameter and creates appropriate trigger point RS-485 telegrams accordingly.
Description	Once a trigger point has been identified by function *Fun-REQ-CCP-Comm-02* above, then this function actually identifies the actions that are to occur as a result of acquiring the specified event trigger. 1) Retrieve Train # from the CCP Parameter table. 2) Extract the specified route station list, trigger point definitions, and content play list definitions from the master schedule data base file, which correspond to the selected route. 3) Continually monitor the "Current Trigger Point" parameter from the CCP parameter table. In the event that the value has changed since the previous read of the parameter, then a new trigger point has been acquired. 4) Search the selected schedule data for entries associated with the current trigger point value. Create an RS-485 trigger event telegram by incorporating the selected data into a valid RS-485 protocol data packet. 5) Set a status flag indicating that a "Trigger Telegram" is pending. 6) Send the telegram when requested by the MASTER PAC.

3.2.3 CNCS

1) After the CNCS completes its boot process it will automatically execute an application called LAUNCHER.EXE which will perform a series of software version checks.

ID	Fun-REQ-CNCS Launcher-01
Function	Validate & Update Software Components
Description	LAUNCHER.EXE must perform the following functions: 1) Access a specified directory within the HDD file structure. 2) Search for the existence of newly uploaded Application (.EXE) files. 3) Perform a CRC check for the newly uploaded Application (.EXE) files. If any new applications fails the CRC check, that file will not be updated) 4) Analyze the associated metadata and retrieve version information. 5) Compare the uploaded File Version information with the File Version information associated with the currently and correspondingly installed components. If the uploaded version(s) are newer, then set flag(s) indicating the necessity to install the specific elements. 6) Determine if there is a new version of LAUNCHER.EXE Application. If so, exit the application while initiating a script to MOVE the updated version to the appropriate location on the HDD and when complete reload the LAUNCHER.EXE application. 7) Clear the LAUNCHER.EXE update flag. 8) MOVE (Update) any remaining flagged Application (.EXE) files from the upload directory of the HDD to its respective installation directory on the CNCS HDD. 9) Exit LAUNCHER.EXE with a parameter to load:CONTENTUPDATE.EXE (Fun-REQ-CNCS Content Update-Initial-01)

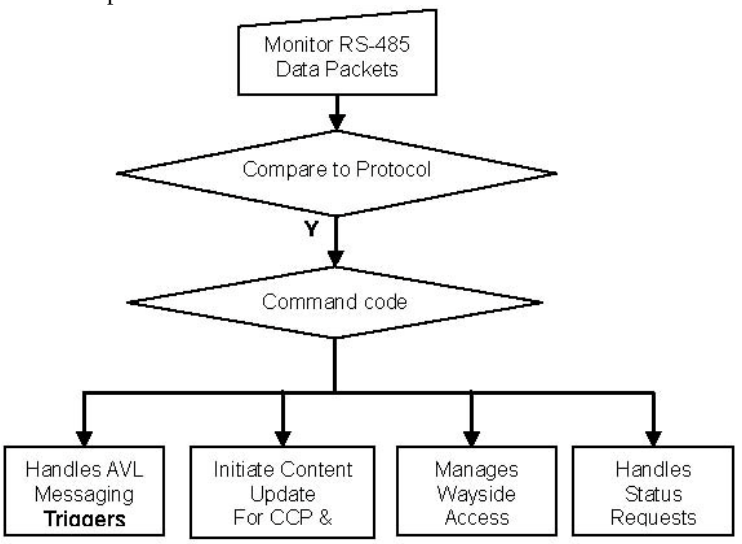
2) Initialize the CNCS-Main program component

ID	**Fun-REQ-CNCS Main-Initial-01**	
Function	Initialize CNCS-MAIN	
Description	This function is utilized to initialize necessary CNCS operating parameters, load and install drivers and communication protocol, and launch all of the various ongoing CNCS component services. 1) Load the default CNCS operating parameters from the onboard nonvolatile memory. 2) Establish an RS-485 Car Line gateway to the User Datagram Protocol (UDP). a) Load and install the required RS-485 Car Line drivers and communication protocol. b) Load the device specific RS-485 Car Line ID. c) Open an RS-485 Car Line port. d) Load the RS-485 device ID table and all relevant packet protocol definitions. 3) Open a UDPServerPort 4) Load and start all CNCS services a) Execute Primary Media Player S/W b) Execute Secondary Media Player S/W c) Execute FTP Data S/W d) Execute MDS Data S/W e) Set Program monitoring timer ON	

3) Monitor CNCS S/W Sub-component Services

ID	**Fun-REQ-CNCS Main-Monitoring-01**
Function	Sub S/W Monitoring
Description	This function monitors the various CNCS S/W component services and evaluates the current operational status of each. 1) Initiate a “Request for Status” update from each S/W component at intervals less than2 seconds. 2) If a specific S/W component does not respond within 5 minute period, then force a conclusion of corresponding S/W process and restart the component.

4) Monitors, Reads and Transmits RS-485 Car Line data

ID	**Fun-REQ-CNCS Main-RS485-01**
Function	Receives & Transmits RS-485-data
Description	This function is used by the CNCS to monitor the RS-485 Car Line activity and enable the required responses. This function performs the following functions: 1) Load RS-485 protocol definitions 2) Monitor the RS-485 port data and compare the received data to the protocol definitions to determine the nature of the data received. 3) Evaluate the specific command sequences specified 4) Assimilate the appropriate controls, protocols, and processes necessary to affect the desired actions. 5) Serve as a gateway to the UDP protocol for delivery of commands to the various sub-component services. 

5) Primary Player Initialization

ID	**Fun-REQ-CNCS PP-Initial-01**
Function	Initialization
Description	Upon the execution of the Primary Player S/W by the main control S/W, this function performs a number of required initialization steps. 1) Open a UDP port. 2) Turn on a timer to indicate that the program is working correctly. 3) Load default parameters 4) Initialize Content Loop functionality.

6) Play CNCS Primary Player Loop Content

ID	**Fun-REQ-CNCS PP-ContentPlay-01**
Function	Play Video Loop Content
Description	The PP S/W plays Advertisements, Public service announcements, or Way finding content in the order specified within the specified loop play list. 1) After loading the content schedule, play the first content file indicated on the schedule. 2) Advertisement content plays at a full size (1280x768) of the LCD as default. 3) When a content is played by the Secondary Player application, the Primary Player will play its content in 1280x668 in the upper zone of the LCD. 4) After completing Play of the first content element, it searches the schedule and plays the next element in the play list if available. 5) It repeats the play list from the beginning once all files registered in the schedule have been played. 6) The content Play Parameters, (including the content file name) are provided by the “Playlist.ini”. 7) Content played via this function will be interrupted in the event that a scheduled AVL trigger is encountered for train service announcements. The loop will resume with the subsequent content element upon completion of the triggered event.

7) Play CNCS Primary Player AVL triggered Content

ID	**Fun-REQ-CNCS PP-RouteInformationPlay-01**
Function	Play AVL route guidance & triggered content
Description	Play route guidance content followed by train operating information which has been indicated in parameters provided by the Main S/W via UDP communication. 1) Analyze parameters (START, APPROACH, ARRIVAL, DOOR, STATION #, and NEXT STATION) received via UDP communication. 2) Stop playing current advertisement content and play the content file specified by the control parameters for the specified station number. 3) Play route guidance content in full screen mode of the LCD. 4) Continue playing the advertisement content file in the content schedule after finishing the route guidance content play. 5) The content Play Parametersare provided by the Main S/W via UDP communication after the Main S/W retrieves the content play parameters from the PAC.

8) CNCS Secondary Player Initialization

ID	**Fun-REQ-CNCS SP-Initial-01**
Function	Initialization
Description	Upon execution of the Secondary Player S/W by the main control S/W, this function performs a number of required initialization steps. 1) Open a UDP port. 2) Load default parameters. 3) Set scrolling area definitions.

9) Plays CNCS-SP AVL triggered Content

ID	**Fun-REQ-CNCS SP-ContentPlay-01**
Function	Plays AVL Triggered Content
Description	Plays the specified AVL triggered content file. 1) Analyze parameters (START, APPROACH, ARRIVAL, DOOR, STATION #, and NEXT STATION) received via UDP communication. 2) Play text-based AVL content in the lower portion(1280x100 from the bottom of the LCD). 3) If the content length fits in the LCD screen, the content will be displayed statically. 4) If the content length is longer than the LCD screen size, the content will be scrolled from the right to the left.

10) Initialize the MDS Data transfer process.

ID	**Fun-REQ-CNCS MDS Data-Initial-01**
Function	Initializing
Description	Executes the following actions to initialize the MDS Data transfer component: 1) Opens the RS-232 port 2) Opens the specifically reserved UDP server port.

11) MDS Data Transfer S/W

ID	**Fun-REQ-CNCS MDS Data-RS232-01**
Function	**Retrieve MDS data packet**
Description	This function retrieves MDS Data from the MDS via the LIC using the RS-232 interface and stores the data to the CNCS HDD. When a specified station code or GPS coordinate is detected, the MDS Data requests to MDS to transmit its data file, and then retrieves it. If approaching a YARD: 1) Retrieve "historical" data from the MDS 2) Store the MDS data into the CNCS HDD (file name = [*Car Number*]_yyyymmddhhmmss [*MDS ID*].bin) If approaching STATION: 1) Retrieve "recent" data from the MDS 2) Store the MDS data into the CNCS HDD (file name = [*Car Number*]_yyyymmddhhmmss [*MDS ID*].bin)

12) Send Time Information

ID	**Fun-REQ-CNCS MDS Data-RS232-02**
Function	Transmit the time information to the MDS via LIC
Description	1) MDS Data monitors the time information of CNCS. 2) Receives a request for time information from LIC 3) Transmits the current time information to LIC

13) Send CS Software Versions

ID	**Fun-REQ-CNCS MDS Data-RS232-03**
Function	Transmit the CS subcomponent S/W version information to the MDS via LIC
Description	1) MDS Data collects and monitors the CS subcomponents SW version information from each CS subcomponents. 2) Receives a request for SW versions from the LIC. 3) Transmits the SW versions to the LIC.

14) CNCS-FTP Initialization Process

ID	Fun-REQ-CNCS FTP-Initial-01
Function	Initializes the FTP protocol and sets operating parameters for communication with the Wayside servers.
Description	The *FTP-Initial-01* function is executed immediately upon launching the FTP S/W component. It must perform the following tasks: 1) Opens a predefined UDP server port. 2) Turns on a Ping timer to specify how often the on-board system Pings the Wayside network for a response. 3) If current location is "Union Station" or a "Yard", then set the timer parameter value to indicate 5 second intervals. 4) Launch the Wayside Timer *Fun-REQ-CNCS FTP-Wayside-01*

15) FTP-Wayside Program Timer

ID	Fun-REQ-CNCS FTP-Wayside-01
Function	The FTP-Wayside Timer carries out a ping test against the wayside periodically to determine the connection status and/or availability.
Description	The Wayside timer attempts to establish communication with the Wayside network by perform a Ping test at parameter defined intervals. 1) Opens a defined UDP port to the CNCS Main application. 2) Set the timer interval based upon a parameter setting 3) Turn on the Wayside timer a) It performs a ping test to the Wayside Network each connection interval to attempt to solicit a response from the Wayside. b) If a valid ping response is received, it tries to connect to the wayside. c) When a valid connection is established it calls the MDS Data Upload function and Content Data Download function. d) If a valid ping response is not received, it retries at the next time interval.

16) FTP-MDS Data Upload

ID	**Fun- REQ-CNCS FTP-Update-01**
Function	Upload MDS Data files from the CNCS to the Wayside
Description	If a train is located in either Station or Yard, it may upload MDS data (error information) as soon as it connects to the Wayside Network. 1) If the next station code is "Union Station" or the GPS coordinate indicates "?? Yard", then the Wayside timer interval is set to 5 seconds so that the train will connect to the Wayside Network ASAP and establish an FTP link. 2) It uploads any MDS data files which are saved on the local CNCS HDD to the Wayside MDS Server via FTP as soon as it has successfully established an FTP session and identified the host.

17) FTP-Content Data Download

ID	**Fun-REQ-CNCS FTP-Update-02**
Function	Download Content Data files from the Wayside to the CNCS
Description	If the Wayside Network connection is valid, carry out the functions listed below: 1) Turn the Wayside Timer off 2) Download various content elements from the wayside. 3) Store each content element on the CNCS HDD in the specified directory location once the download has been successfully completed. 4) If there is no content available or necessary to download from the Wayside, or if the Wayside Network connection is lost, then the FTP session will be terminated and the Wayside Timer will be set to the "On" state.

18) CNCS Content Update

ID	**Fun-REQ-CNCS ContentUpdate-Initial-01**
Function	Initializes the Content Update functional component
Description	Upon execution of Content update S/W component, it performs a number initialization processes. 1) Load default operating parameters 2) Opens a UDP port to the Main application 3) Loads set value of RS-485 4) Opens RS-485 port

19) Transfer PRIMARY Authority regarding RS-485 Communication from the PAC to the CNCS

ID	**Fun-REQ-CNCS ContentUpdate-RS485-01**
Function	Set / Define RS-485 data Transmit/Receive Mode
Description	In order to effective facilitate transferring large amounts of data as required by the Content Update function, it is necessary to temporarily pass RS-485 Car Line control from the Primary PAC to the CNCS. At the completion of the Content Update process control is returned to the PAC for normal operations. 1) Request primary authority transfer from PAC to CNCS a) When the Primary PAC requests a status update from the CNCS, the CNCS will respond with a request for authority transfer so that it can manage the RS-485 lines to perform the Content updates. b) Once the Master PAC has granted the CNCS master authority over the RS-485 Car Lines, then this function will call the “Content Update” function. 2) Received data packet passes checksum analysis 3) Upon return from the Data Update function, RS-485 control is passed back to the Primary PAC.

20) Content Update Call Command:

ID	**Fun-REQ-CNCS ContentUpdate-Update-01**
Function	Calls the “Content Update” function with the appropriate command parameters
Description	Calls the Content Update S/W with different parameters in order to select the appropriate update function based on the specific devices and type of content being updated. This function updates content for many of the CS sub-systems including the following: 1) Call the “CCP-Update” function if a new version of the train operating schedule has been downloaded to the “Update” directory of the CNCS HDD. 2) Call the “MP3-Update” function if new AMF(Audio Message Format) files have been downloaded to the “Update” directory of the CNCS HDD. 3) Call the “LED-Update” new LED HEX Files have been downloaded to the “Update” directory of the CNCS HDD. 4) After all updates have been completed, return primary control of the RS-485 Car Line back over to the Primary PAC. 5) Close the RS-485 port. 6) Close the Content Update S/W and return to the Main Control S/W upon completion of the update process.

21) Update Train Schedule Database

ID	Fun-REQ-CNCS ContentUpdate-Update-02
Function	CCP-Update function
Description	This function Updates the Master Train Schedule components in the CCP as necessary. 1) Transfer ready command indicating update to the CCP is pending and will be performed by the CNCS in bulk data mode. 2) Confirm status for ready to update in each CCP. Transmit updated Train Operating Schedule data. 3) Transmit updated GPS Trigger Point data. 4) Transmit updated Station Naming data. 5) Transmit updated Route Naming data. 6) Transmit updated Holiday Schedule data. 7) Validate the data transferred to the CCP after the Data packet transfer process has completed. 8) Return to the Content Update function regardless of the completion and validity status.

22) Update automated PA message file to SD memory in PAC via USB interface.

ID	Fun-REQ-CNCS ContentUpdate-Update-03
Function	MP3_Update function
Description	Update the AMF(Audio Message Format) file library via USB communication to the PAC's MP3 Player USB memory located in CRA. 1) Transfer ready command for USB update in PAC. 2) The PAC which receives the ready request command responds with confirmation of its ready status. 3) The CNCS transfers the AMF(Automated Message File) files from the CNCS HDD to the USB memory module. 4) Check the update status to ensure proper completion of the process. 5) Send an "Update Complete" command to the PAC. 6) The PAC disconnects the USB connection between the CNCS and the PAC's USB memory. 7) This function then returns to the "Content Update" function .

23) Update LED signs with LED statement data using RS-485 communication.

ID	Fun-REQ-CNCS ContentUpdate-Update-04
Function	Updates LED Content Data using RS-485
Description	Updates both the image based and text based LED content elements 1) LED Content update is carried out in order of FDI, SDI and PII. 2) Send a "Get Ready" command to each sign in the specified category, and record their "Ready" responses. 3) Transfer the new LED Content data to the LED signs by page(256 byte) 4) If communication errors have occurred, then repeat the process up to two times for each of the specific signs which indicated an erroneous transfer. 5) After all signs in the category have been updated successfully, then go on to the next category and repeat the process. 6) This function then returns to the "Content Update" function for completion of the LED display update process.

3.2.4 FDI

1) FDI Initialization Process

ID	Fun-REQ-FDI-Initial-01
Function	Initialize the FDI main board
Description	1) Initialize the FDI operating S/W. 2) Set the FDI RS-485 DEVICE_ID after checking the Address Signal. 3) Open RS-485 communication port

2) Respond to FDI Status Request

ID	Fun-REQ-FDI-Comm-01
Function	Responds to requests for status data from the PAC indicating the operational status of the FDI.
Description	Responds to the Primary PAC's request for status update via the Car Line RS-485 by sending the current status data of the device. The Master PAC then polls each Primary PAC and consolidates the status data in the CCP for review.

3) Display FDI Content

ID	Fun-REQ-FDI-Display-01
Function	Display the appropriate information on the FDI as indicated by the current train operating parameters.
Description	1) Receives the Train Route code & related information from the PAC 2) Load LED Image Data relative to the station, Route Number, and Destination Content into internal memory 3) Display image data from internal memory

4) FDI Content Update Management

ID	Fun-REQ-FDI-Update-01
Function	Receives LED Image Data from CNCS and updates the content library stored in local FDI non-volatile memory. Does not delete previous content until FDI is completely updated.
Description	1) Receive a command to update LED Image data from CNCS 2) Switch to content update mode 3) Receive LED Image data packets from CNCS via RS-485 4) Store received data in local FDI memory 5) Validate integrity of transferred content elements 6) Delete replaced and/or expired content elements 7) Return to normal operating mode

3.2.5 SDI

1) SDI Initialization Process.

ID	Fun-REQ-SDI-Initial-01
Function	Initialize the SDI main board
Description	1) Initialize the SDI operating S/W. 2) Set SDI RS-485 DEVICE_ID after checking the Address Signal. 3) Open the RS-485 communication port

2) Respond to Request for SDI status data

ID	Fun-REQ-SDI-Comm-01
Function	Responds to requests for status data from the PAC indicating the operational status of the SDI.
Description	Responds to the Primary PAC's request for status update via the Car Line RS-485 by sending the current status data of the device. The Master PAC then polls each Primary PAC and consolidates the status data in the CCP for review.

3) Display SDI Content

ID	Fun-REQ-SDI-Display-01
Function	Display the appropriate information on the SDI as indicated by the current train operating parameters.
Description	1) Receives the Train Route code & related information from the PAC 2) Load LED Image Data relative to the station, Route Number and Destination Content into internal memory. 3) Display image data from internal memory

4) SDI Content Update Management

ID	Fun-REQ-SDI-Update-01
Function	Receives LED Image Data from CNCS and updates the content library stored in local SDI non-volatile memory. Does not delete previous content until SDI is completely updated.
Description	1) Receive a command to update LED Image data from CNCS 2) Switch to content update mode 3) Receive LED Image data packets from CNCS via RS-485 4) Store received data in local SDI memory 5) Validate integrity of transferred content elements 6) Delete replaced and/or expired content elements 7) Return to normal operating mode

3.2.6 PII

1) PII Board Initialization Process

ID	Fun-REQ-PII-Initial-01
Function	Initialize PII main board
Description	1) Initialize the PII operating S/W. 2) Set PII RS-485 DEVICE_ID after checking the Address Signal. 3) Open RS-485 communication port

2) Response to PII Status Request

ID	Fun-REQ-PII-Comm-01
Function	Responds to requests for status data from the PAC indicating the operational status of the PII.
Description	Responds to the Primary PAC's request for status update via the Car Line RS-485 by sending the current status data of the device. The Master PAC then polls each Primary PAC and consolidates the status data in the CCP for review.

3) Display AVL Station Arrival and Departure Messages

ID	Fun-REQ-PII-Display-01
Function	Display text font based AVL Station Arrival and Departure messages.
Description	This function provides for the ability to display the standard AVL arrival and departure messages, (BMP/HEX Based), which are designated in the master route schedule for the currently selected train number. 1) When a trigger point is discovered by function Fun-REQ-PII-Comm-01 it initiates a start command including parameters which specify the specific content element to be displayed, the method of display, and the duration. 2) Execute the content play command by loading the specified content data from the PII non-volatile memory into the display memory area. 3) Maintain the displayed image for the specified period of time, or until terminated by a subsequent play command or cancel command. 4) Exit the routine

4) PII Content Update management

ID	Fun-REQ-PII-Update-01
Function	Receives LED Image Data from CNCS and updates the content library stored in local PII non-volatile memory. Does not delete previous content until PII is completely updated.
Description	1) Receive a command to update LED Image data from CNCS 2) Switch to content update mode 3) Receive LED Image data packets from CNCS via RS-485 4) Store received data in local PII memory 5) Validate integrity of transferred content elements 6) Delete replaced and/or expired content elements 7) Return to normal operating mode

3.2.7 LIC

1) LIC Device Initialization Process

ID	Fun-REQ-LIC MAIN-Initial-01
Function	Initialize the LIC Board at Power-On
Description	1) Initialize processor chip after power-on 2) Set LIC_ID after checking the LIC_ID signal. 3) Check LIC-LON S/W status.

2) Retrieves device fault data from the MDS via the LIC and transmits it to the CNCS.

ID	Fun-REQ-LIC MAIN -Comm-01
Function	Receives device fault data from the LIC and writes it to a file on the CNCS
Description	This function retrieves MDS Data from the MDS via the LonWorks interface and forwards the data stream to the CNCS-MDS S/W component via the RS-232 interface. The specific data retrieved will depend upon the location of the vehicle as determined by the CCP route processor. If however the vehicle is at a “YARD” then all of the available MDS data must be retrieved. The appropriate trigger data to accomplish this will be forwarded by the CCP to the CNCS main S/W via the RS-485 Car Line. The data will then be incorporated into operating parameters and sent to the CNCS-MDS Data Transfer S/W component via UDP. The CNCS-MDS S/W component will then issue an appropriate request for response to this LIC S/W component. 1) Receives a request for train fault data from CNCS 2) Requests either “Recent” or “Historical” fault data via LonWorks Communication provided by the LIC using the LonWorks S/W component 3) Receives fault data stream from the MDS and transmits it in real-time to the CNCNS using the RS-232 communications channel.

3) Receives CS fault data for each sub system from the PAC and transfers it via the LIC to the MDS.

ID	Fun-REQ-LIC MAIN -Comm-02
Function	The LIC Main Control S/W transfers device fault data received from the PAC via RS-485 to the LIC and then forwards it to the MDS.
Description	1) The LIC Main Control S/W monitors the CS fault data from PAC. 2) MDS requests for the CS fault data to LIC Main Control S/W via the LonWorks interface. 3) LIC Main Control S/W transmits the current CS fault status to MDS device via LonWorks interface.

4) Send LIC Status Data to the PAC.

ID	Fun-REQ-LIC MAIN -Comm-03
Function	Transmit the LIC operational status data to the PAC
Description	1) Receives a request for LIC status from the PAC. 2) Transmits the LIC's current operational status to the PAC.

5) Send Time Information to the MDS

ID	Fun-REQ-LIC MAIN -Comm-04
Function	Transmit the time information to the MDS
Description	1) LIC-MAIN monitors the current time information provided by CNCS MDS Data. 2) Receives a request for time information from the MDS. 3) Transmits the current time information to the MDS.

6) Send CS Software Versions to the MDS

ID	Fun-REQ-LIC MAIN -Comm-05
Function	Transmit the CS subcomponent S/W version information to the MDS
Description	1) LIC-MAIN monitors the CS subcomponents SW version information provided by CNCS MDS Data application. 2) Receives a request for SW versions from the MDS. 3) Transmits the SW versions to the MDS.

7) Device Initialization

ID	Fun-REQ-LIC LON-Initiali-01
Function	Initializing
Description	1) Initialize processor chip upon power-on. 2) Send LIC Status to Lon Network.

8) Provide a Gateway between (Convert Data Format and Protocol) between the LonWorks Protocol and the LIC Main Control S/W for transmitting & receiving data.

ID	Fun-REQ-LIC LON-Comm-01
Function	This S/W Component (LIC LonWorks) functions in conjunction with the previously defined S/W Component (LIC LonworksMain Control) to provide for a Gateway/Bridge between the LonWorks data network and the CS device protocols.
Description	1) Convert data from LonWorks Protocol and write to internal memory in a format that can be read via the LIC Main S/W and then forwarded to the CS Devices via either RS-232 or RS-485. 2) These two S/W components work bi-directionally

3.3 Performance Requirements

- VTX

H/W Circuit of VTC should receive VTX_ID Address Signal and set its own ID at the same time of a power permission. It should not impact on RS-485 Interface when ID is not set.
Check if the video signal received from CNCS is existent and respond the status of H/W via PAC. The signal received by VTC in PAC-MA should be bypassed if video signals cannot be transmitted to VRX due to device failure or no video signal input to VTX.

- VRX

H/W Circuit of VRX should receive VRX_ID Address Signal and set its own ID at the same time of a power permission. It should not impact on RS-485 Interface when ID is not set.
Check if the video signal received from VTX or VRX is existent and respond the status of H/W via PAC. The signal received should be bypassed if video signals cannot be transmitted to VRX due to the device failure of VRX.

- FDI

FDI should receive FDI_ID Address Signal and set its own ID at the same time of a power permission. It should not impact on RS-485 Interface when ID is not set. It should response the status via PAC after ID setting. The image to be displayed on LED module should be stored in a specific image memory. It should be applicable after the update is completed when the image update via CNCS. The image of the former version should be normally displayed when the update is not completed.

- SDI

SDI should receive SDI_ID Address Signal and set its own ID at the same time of a power permission. It should not impact on RS-485 Interface when ID is not set. It should response the status via PAC after ID setting. The image to be displayed on LED module should be stored in a specific image memory. It should be applicable after the update is completed when the image update via CNCS. The image of the former version should be normally displayed when the update is not completed.

- PII

PII should receive PII_ID Address Signal and set its own ID at the same time of a power permission. It should not impact on RS-485 Interface when ID is not set. It should response the status via PAC after ID setting. The image to be displayed on LED module should be stored in a specific image memory. It should be applicable after the update is completed when the image update via CNCS. The image of the former version should be normally displayed when the update is not completed.

- PAC

PAC is the main device which manages the RS-485 communication within the Communication system. Every device which participates in the communication should not hold the RS-485 Enable on the communication line by malfunction. Train will be separated physically into 2 by the married car composition. PAC shows malfunction in the announcement, communication, and vehicle consist information collection, if the RS-485 which connects between the vehicle and another vehicle, and train line RS-485 gets disconnected.

- DPO

DPO receives tone signal and delivers command with the tone signal. All DPO, DPH, VOIP, and PEI of vehicle which has been double-head connected should be connected on the signal line which delivers tone. The ID of the DPO should be distinguished with other DPOs in the same vehicle, and accurate operation is available when signal output address line has been connected to be distinguished.

- DPH

DPH receives tone signal and delivers command with the tone signal. All DPO, DPH, VOIP, and PEI of vehicle which has been double-head connected should be connected on the signal line which delivers tone. The ID of the DPH should be distinguished with other DPHs in the same vehicle, and accurate operation is available when signal output address line has been connected to be distinguished.

- PEI

PEI receives tone signal and delivers command with the tone signal. All DPO, DPH, VOIP, and PEI of vehicle which has been double-head connected should be connected on the signal line which delivers tone. The ID of the PEI should be distinguished with other PEIs in the same vehicle, and accurate operation is available when signal output address line has been connected to be distinguished. PEIs should be set not to occur howling as the PEI locates close to microphone and speaker.

3.4 Logical Database Requirements

None.

3.5 Design Constraints

3.5.1 General Design Constraints

- The amount of memory capacity used by the installed S/W shouldn't be over 80% of the entire memory capacity.
- 32bit STR7 Chip is used for Microprocessor.
- The application S/W operated on CNCS is automatically updated by wireless network.
- Windows XP Embedded based Application S/W shall be automatically implemented when Operating System is booted.
- The Destination Sign System shall utilize Global Positioning System (GPS) technology to provide location information.
- Route display system should use GPS to provide location information.
- Communication System should be completely independent of communication systems of DEVER traffic system(bus/LRV/)
- When Communication System does not used, the sounds should be muted.
- PID should display 20 characters by 4 rows simultaneously, the height of character should be about 2 inches, and PID should display graphics and images.
- The updates of location database and system software should be carried out using RS-232 serial port and IBM compatible portable computer.
- The GPS obtained time SHALL be used for all system clocks on the train in UTC

3.5.2 Standard Compliance

Operational manual for end user shall follow the guide-line of IEEE STD 1063. - S/W Source code shall comply with industry standard procedures .

3.6 Software System Attributes

3.6.1 Reliability

- The sub S/W in the CNCS is monitored every 1second by Main control S/W. If any particular sub S/W does not respond for 5 second, it shall be re-operated after the relevant S/W is finished by Main control S/W.
- Each device which has built in MCU uses Watch Dog function in the Chip to recover it by resetting MCU when device operated abnormally.
- PAC reduces probability of malfunction by using dual channel to get over outside noise when receiving intercom control signal.

3.6.2 Availability

- PAC is the most important device for Communication System and it is dualized as CRA. If Master PAC is not operating for 3 seconds, it should automatically operate slave PAC as Master

3.6.3 Security

- On-Board CS Software can only be accessed or updated via a) Password protected PTU, or b) secure automated Wayside updates controlled by DENVER RTD and incorporating version control techniques.

3.6.4 Maintainability

- For the effective maintenance of source code, internally defined S/W coding guide shall be followed and comment display rate by module shall be over 50%.

3.6.5 Portability

- Application S/W operated on CNCS is portable to Windows XP OS.